

Finite reservoirs at phase coexistence: full-state accuracy and phase correlations

Abstract

We determine when a finite thermal reservoir reproduces the complete canonical law of a coexisting subsystem, allowing optimization of the composite total energy. For a reservoir density of states proportional to U^{c_N} , a weak energy limit on scale b_N with at least three support points requires and permits total-variation convergence exactly when $c_N \gg b_N^2$. Two energy phases are exceptional: a gap Δ_N and nondegenerate fluctuation scale s_N require $c_N \gg \Delta_N s_N$, with matching sufficiency under explicit positive phase-tail bounds. We verify the resulting $N^{3/2}$ threshold for mean-field and sufficiently-large- q short-range Potts systems. Positive contour estimates provide the needed short-range energy moments, including for plain spins. Under stronger phase-tail and exceptional-mass assumptions, at the $N^{3/2}$ boundary we derive the full-state error and optimize it over every composite energy; an optimum can discard a phase. For two independent copies sharing a bath, $N \ll c_N \ll N^2$ can preserve both balanced canonical marginals while enforcing opposite phases, joint total-variation error 1/2, and one bit of microscopic mutual information. Joint canonical accuracy instead requires $c_N \gg N^2$. Exact Gaussian geometry, smooth-reservoir extensions, and a stated capillarity model clarify which conclusions depend on phase support, reservoir curvature, or interfacial tails. Exact finite-system occupation calculations illustrate the physical thresholds.

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1 Introduction

A finite reservoir changes the probabilities of subsystem energies. Away from a phase transition, an energy window of width proportional to $\sqrt{N}$ often contains nearly all canonical probability. At first-order coexistence, two or more such windows can be separated by an energy difference proportional to N . A reservoir that is adequate within one window can then change the relative

phase populations, the fluctuations inside the phases, or both. The question addressed here is how large a physical thermal reservoir must be to reproduce the complete canonical law of an interacting subsystem, after its total energy has been calibrated as favorably as possible.

The question arises wherever a small system exchanges energy with a bath that is not much larger than itself. Finite-bath and Gaussian ensemble Monte Carlo schemes deliberately use such baths to sample first-order transitions [6, 20]; nanoscale calorimetry of clusters and confined fluids compares a finite sample with a finite thermal environment; and a subsystem of a closed classical or quantum system thermalizes only to the accuracy that the remainder can supply [30]. In each case a single number, the bath heat capacity, is expected to control the error, and the practical question is how it must scale with the subsystem size at coexistence.

We use total variation, $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$. It controls the error in every bounded measurement of the subsystem, including its phase label and its fluctuation-scale observables. Agreement of thermodynamic potentials, energy density, or a specified finite set of averages does not in general establish full-law TV convergence. Conversely, convergence of unbounded averages requires additional integrability, as discussed in Section 2. Full-law TV is also a different requirement from a vanishing unscaled Euclidean distance between broad energy histograms. Because our reservoir likelihood depends only on subsystem energy, energy-law TV equals full-state TV exactly. Energy is therefore sufficient for computing the strong error without replacing the microscopic system by an energy-only model.

The reservoir has density of states $\omega_B(U) \propto U^{c_N} \mathbf{1}_{\{U>0\}}$. The exponent $c_N > 0$ is its surface-entropy heat capacity in units of k_B ; Section 2 states the finite-system entropy convention and its distinction from the canonical heat capacity. The subsystem Hamiltonian and target canonical inverse temperature are fixed in each comparison. We optimize only the total energy of the isolated composite. Weak interaction permits energy exchange but contributes no interaction energy to this equilibrium marginal.

1.1 The mechanism in two lines

The subsystem sees the bath through the log likelihood $\beta e + c_N \log(\mathcal{E}_N - e)$, a concave function of the subsystem energy e with curvature of order $1/c_N$. Choosing the total energy $\mathcal{E}_N$ so that this function takes equal values at two phase centers separated by Δ_N leaves endpoint slopes of order Δ_N/c_N . Across a phase of energy width s_N such a slope tilts the log weight by $\Delta_N s_N/c_N$, so the full law survives only if $c_N \gg \Delta_N s_N$; for an extensive latent heat and $\sqrt{N}$ fluctuations this is $c_N \gg N^{3/2}$. If instead three energies on a common scale b_N must all receive equal weight, no choice of $\mathcal{E}_N$ helps: a concave function of curvature $1/c_N$ cannot be flat at three points spaced by b_N unless $b_N^2/c_N \rightarrow 0$. The rest of the paper makes both statements sharp, in both directions, and verifies their hypotheses for interacting spin models.

1.2 Results

The required capacity depends on the energy support that must be preserved. If $(E_N - a_N)/b_N$ has a weak limit containing at least three support points, the optimal condition is $c_N \gg b_N^2$. This theorem requires no density, central limit theorem, or moment bound. Here $x_N \gg y_N$ means $x_N/y_N \rightarrow \infty$. For a single phase with a nondegenerate Gaussian limit on scale $\sqrt{N}$, it gives $c_N \gg N$. For three distinct macroscopic energy phases it gives $c_N \gg N^2$.

Two energy phases admit the secant calibration just described. We prove the condition $c_N \gg \Delta_N s_N$ necessary when at least one phase has a nondegenerate weak fluctuation limit, and sufficient under explicit positive-phase exponential-moment and exceptional-mass bounds. Only two distinct

Table 1: Optimal physical reservoir scales. Each convergence statement concerns full-state TV and permits total-energy calibration. The table summarizes asymptotic theorems under the indicated hypotheses; it is not a universal finite-size error bound.

Canonical energy structure	Required capacity	Hypotheses and result
One ordinary fluctuation window	$c_N \gg N$	Nondegenerate weak Gaussian limit on scale $\sqrt{N}$; Theorem 3.1
Two extensive energy phases	$c_N \gg N^{3/2}$	Positive phases, one nondegenerate $\sqrt{N}$ limit, controlled phase tails and exceptional mass; Corollary 3.6
At least three extensive energy phases	$c_N \gg N^2$	Weak macroscopic limit with at least three support points; Theorem 3.1
Two independent two-phase copies, joint target	$c_N \gg N^2$	Three support points for the total energy; Corollary 5.3

energies are involved: several symmetry-related ordered phases at the same energy do not create a three-energy obstruction.

The two-phase theorem is realized by both a three-state Curie–Weiss–Potts Hamiltonian and a short-range Potts Hamiltonian with a fixed large number of colors, in every fixed spatial dimension $d \geq 2$. Quadratic kinetic coordinates are optional in both models. For the short-range system, the main difficulty is controlling the positive probability carried by interfacial configurations after reservoir reweighting. We obtain phase-energy exponential moments from positive restricted contour partition functions, then transfer them from occupied bonds to the exchanged spin energy. A signed finite-volume partition error alone would not establish this control.

For these microscopic examples, let w_-, w_+ denote the limiting phase probabilities. The subsystem’s canonical heat capacity is itself of order N^2 at coexistence:

$$\frac{C_{S,\text{can}}}{k_B N^2} = \frac{\beta_N^2 \text{Var}_{P_N}(E_N)}{N^2} \longrightarrow \beta^2 w_- w_+ \ell^2, \quad \ell = \lim_N \Delta_N / N. \quad (1.1)$$

Indeed, the bounded spin energy per particle converges to the two-atom law, so its first two moments converge. Optional independent kinetic energy with Gamma shape aN adds only aN/β_N^2 to the variance; $a = 0$ denotes no kinetic sector. The proved phase moments bound within-phase variances by order N ; the leading N^2 contribution instead comes from the separation of the phases. Secant calibration compensates their relative bias, permitting full-law accuracy with $N^{3/2} \ll c_N \ll N^2$, even though the bath heat capacity is then smaller than this full canonical coexistence heat capacity. This is the physical content of the $N^{3/2}$ scale: a bath need not be able to absorb the latent heat at a fixed temperature in order to reproduce the coexisting law exactly.

A shared bath reveals a second distinction. For two otherwise independent copies, a capacity in the window $N \ll c_N \ll N^2$ can suppress equal-phase pairs while retaining ordinary fluctuations in opposite-phase pairs. At balanced canonical phase weights, both full subsystem marginals converge to their canonical laws, whereas joint TV tends to $1/2$ and the full microscopic mutual information tends to $\log 2$. The reference temperature needed to balance the phases is specified and used consistently by the bath; for the spin models it is a shift of order $1/N$ from the coexistence temperature. Joint canonical accuracy requires the larger N^2 scale because the sum of two two-phase energies has three macroscopic support points.

At the boundary $c_N \sim \gamma N^{3/2}$, stronger phase-tail integrability and exceptional-mass suppression allow us to determine the finite Gaussian phase tilts, their integrated probabilities, and the full TV

error. These hypotheses hold for every fixed $\gamma > 0$ in the mean-field model and the short-range model in dimensions above two; in two dimensions our proof gives them for sufficiently large γ . We optimize this error over every admissible composite energy, including choices that discard a phase. This matters operationally: minimizing one scalar distance can favor losing a phase rather than retaining its probability with distorted fluctuations. Exact multivariate Gaussian models in Appendix B show a related geometric constraint: the phase points that can survive optimal quadratic reweighting are governed by empty-sphere contact sets.

1.3 Calibration sensitivity

The secant calibration must be accurate. In the regime $c_N \gg \Delta_N$, shifting $\mathcal{E}_N$ by δ changes the difference of the two center log weights by approximately $\beta_N^2 \Delta_N \delta / c_N$, so the phase odds change by a factor of order one when $|\delta| \asymp c_N / (\beta_N^2 \Delta_N)$. Preserving the limiting phase weights of the calibrated law instead requires $\delta = o(c_N / (\beta_N^2 \Delta_N))$. At either center the bath energy is $U_{B,i,N} = \mathcal{E}_N - e_{i,N} \sim c_N / \beta_N$. Thus, for the microscopic models with $\Delta_N \asymp N$, relative bath-energy errors of order $1/N$ change the phase odds by order one; vanishing changes require relative errors $o(1/N)$. The same holds for relative errors in the bath temperature at either center. The canonical ensemble has exactly the same sensitivity at coexistence: a change of order $1/N$ in β multiplies the phase odds by a factor of order one. The requirement is therefore a property of the target law, not of the finite bath. Section 6 quantifies the effect of calibration errors of order $\sqrt{N}$ at the $N^{3/2}$ boundary, where they change the limiting phase weights but not the fluctuation tilts.

1.4 Relation to earlier work

Finite-reservoir ensembles and their fluctuation corrections are established. Challa and Hetherington introduced Gaussian ensembles as a controllable interpolation between familiar ensembles and demonstrated changes to first-order energy histograms [7, 6]. The two-Gaussian coexistence description of Challa, Landau, and Binder [8] already separates an extensive latent-energy gap from ordinary within-phase widths, and their later review covers finite-size and finite-bath approaches [9]. Power-law subsystem laws from finite constant-capacity reservoirs are standard [5], and the surface-versus-volume entropy convention for the bath is the one compared in [21, 17].

Strong-distance bath bounds also exist. Riera, Gogolin, and Eisert [30] control a reduced microcanonical-shell state in trace distance by the nonlinear remainder of the bath entropy; in their bath example a sufficient size is proportional to the square of the subsystem energy range, which in a commuting setting is a TV bound of the N^2 type. Sharp growing-block conditional limit theorems [16] treat a regular exponential-family conditioning problem rather than the full phase mixture of an interacting subsystem coupled to a separate reservoir. Finite-bath Potts comparisons by Griffin, Matty, and Swendsen [20] are the closest numerical precedent; they measure agreement by an unweighted Euclidean histogram norm and optimize a comparison temperature, and Appendix C gives an exact example in which that norm vanishes while full-state TV does not. Generalized Gaussian ensembles recover nonconcave entropy and macrostate equivalence at fixed penalty [13, 14], and squeezed ensembles select first-order states with finite-size conversion formulas [31]; both are thermodynamic-limit questions at fixed penalty rather than complete-law asymptotics at shrinking curvature. For shared baths, ensemble dependence of fluctuations and correlations is classical [25, 27]; opposite-phase assignments in two weakly coupled identical systems were obtained by Ramírez-Hernández, Larralde, and Leyvraz [29]; and ensemble-dependent shared information, including order-one contributions from degenerate phases, was studied by Cohen, Rittenberg, and Sadhu [10].

Against this background, the contributions of the present paper are the following. First, sharp necessity: for the exact power-law bath and every total-energy calibration, a weak energy limit with three support points forces $c_N \gg b_N^2$, and two phases with a nondegenerate fluctuation limit force $c_N \gg \Delta_N s_N$. Second, the optimized two-phase scale $N^{3/2}$ is attained, below both the energy-range-squared scale and the coexistence heat capacity, under positive phase-tail hypotheses that we state explicitly. Third, these hypotheses are verified for mean-field and short-range Potts Hamiltonians, the latter through positive contour measures and a bond-to-energy transfer. Fourth, the boundary law at $c_N \asymp N^{3/2}$ is computed and optimized over every composite energy, including phase abandonment. Fifth, under a shared bath the full marginal, joint, and information limits are proved in a specified capacity window.

1.5 Organization

Section 2 derives the exact reservoir likelihood and calibrations. Section 3 proves the weak-support and two-phase criteria. Section 4 gives microscopic realizations and exact finite-system formulas; Appendix A contains the contour proof. Section 5 treats joint statistics under a shared bath, and Section 6 gives smooth-bath extensions, the optimized physical boundary, and the exact interior gain of the secant bath. Section 7 presents reproducible deterministic calculations. The exact Gaussian geometry, the capillarity and accuracy diagnostics, and the shared-bath extensions to fixed phase counts and linear capacity are collected in Appendices B, C, and D.

2 Physical reservoir and accuracy criterion

2.1 Canonical reference and the exact composite marginal

Let $(\Omega_N, \mathcal{F}_N, \mu_N)$ describe the subsystem states, with measurable finite energy $E_N : \Omega_N \rightarrow \mathbb{R}$. The reference measure may be counting measure or a classical phase-space measure. At the specified inverse temperature $\beta_N > 0$, assume

$$P_N(dx) = \frac{e^{-\beta_N E_N(x)}}{Z_N} \mu_N(dx), \quad 0 < Z_N < \infty. \quad (2.1)$$

Throughout, $\beta_N \rightarrow \beta \in (0, \infty)$; a fixed temperature is included. Allowing this specified sequence is useful for finite-size equal-weight coexistence. The target law is chosen before the reservoir is tuned.

We neglect the interaction energy between subsystem and reservoir. The reservoir has the exact surface density of states

$$\omega_{B,N}(U) = A_N U^{c_N} \mathbf{1}_{\{U>0\}}, \quad A_N > 0, \quad c_N > 0. \quad (2.2)$$

Here “surface” means the density of states at an energy, rather than its integral up to that energy. Integrating a microcanonical energy shell for the composite gives, at composite energy $\mathcal{E}_N$, the subsystem law

$$Q_N(dx) = \frac{L_N(E_N(x))}{Z_N} P_N(dx), \quad L_N(e) = \begin{cases} e^{\beta_N e} (\mathcal{E}_N - e)^{c_N}, & e < \mathcal{E}_N, \\ 0, & e \geq \mathcal{E}_N, \end{cases} \quad Z_N = \mathbb{E}_{P_N} L_N(E_N). \quad (2.3)$$

The bath exponent c_N is prescribed. Only $\mathcal{E}_N$ is adjustable. For every finite $\mathcal{E}_N$, the function L_N is bounded: it is continuous below the cutoff and tends to zero both at the cutoff and as $e \rightarrow -\infty$. Thus $Z_N < \infty$ automatically. The total energy is admissible when $P_N(E_N < \mathcal{E}_N) > 0$, which is exactly the condition $Z_N > 0$.

To specify the heat-capacity convention, write $S_{B,N}(U) = k_B \log \omega_{B,N}(U)$, with a fixed reference energy absorbed in the additive constant. Its temperature T_B satisfies $1/T_B = \partial_U S_{B,N} = k_B c_N/U$. Hence

$$\beta_B = \frac{1}{k_B T_B} = \frac{c_N}{U}, \quad C_B = \frac{dU}{dT_B} = k_B c_N. \quad (2.4)$$

For $f_N > 2$ independent classical quadratic degrees of freedom, $c_N = f_N/2 - 1$. At a canonical inverse temperature β_{can} , the same reservoir has an energy density proportional to $U^{c_N} e^{-\beta_{\text{can}} U} \mathbf{1}_{\{U>0\}}$, with mean energy $(c_N + 1)/\beta_{\text{can}}$ and canonical heat capacity $k_B(c_N + 1)$. This finite offset follows from the chosen surface convention and does not affect the diverging scales proved below. The theorems concern the specified density of states and exponent; they do not require a choice between the competing surface and volume definitions of microcanonical entropy compared by Hilbert, Hänggi, and Dunkel [21, 17]. No quadratic expansion of the bath entropy is assumed in (2.3).

2.2 Full-state and energy-law accuracy coincide

For probability measures P, Q on the same space, use

$$d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2} \int |dP - dQ|. \quad (2.5)$$

The last integral uses any common dominating measure. Let P_N^E, Q_N^E be the pushforwards by E_N .

Lemma 2.1 (Energy is sufficient for reservoir reweighting). *For the exact marginal (2.3),*

$$d_{\text{TV}}(Q_N, P_N) = d_{\text{TV}}(Q_N^E, P_N^E) = \frac{1}{2} \mathbb{E}_{P_N} \left| \frac{L_N(E_N)}{Z_N} - 1 \right|. \quad (2.6)$$

Proof. The Radon–Nikodym derivative on the state space is $L_N(E_N)/Z_N$. Its pushforward derivative is the same function of energy: for every bounded measurable f ,

$$\mathbb{E}_{Q_N} f(E_N) = \frac{1}{Z_N} \int f(e) L_N(e) P_N^E(de).$$

The two L^1 formulas for total variation are therefore identical. $\square$

We consequently work with energy laws without losing any information about this full-state distance. Matching a mean energy, a pressure, or a few local observables need not determine this distance. Conversely, total variation controls uniformly bounded observables; convergence of expectations of unbounded observables requires additional tail control, such as uniform integrability under both laws. The equality in Lemma 2.1 depends on the energy-only likelihood and need not hold when coupling changes the subsystem Hamiltonian.

Lemma 2.2 (Normalization criterion). *Suppose $W_N \geq 0$ and $\mathbb{E}_{P_N} |W_N - 1| \rightarrow 0$. Then $\mathbb{E}_{P_N} W_N \rightarrow 1$, and the probability law with derivative $W_N/\mathbb{E}_{P_N} W_N$ converges to P_N in total variation. Conversely, total-variation convergence of laws with derivatives R_N implies $R_N \rightarrow 1$ in P_N -probability. In particular, if $0 \leq W_N \leq 1$ and $W_N \rightarrow 1$ in probability, the first assertion applies.*

Proof. Writing $z_N = \mathbb{E} W_N$, we have $|z_N - 1| \leq \mathbb{E} |W_N - 1|$ and

$$\frac{1}{2} \mathbb{E} \left| \frac{W_N}{z_N} - 1 \right| \leq \frac{\mathbb{E} |W_N - 1| + |z_N - 1|}{2z_N}.$$

The converse follows from Markov's inequality and (2.5). The last assertion follows from boundedness and convergence in probability. $\square$

All asymptotic scales refer to fixed energy units. We write $u_N \gg v_N$ when $u_N/v_N \rightarrow \infty$, and $u_N \asymp v_N$ when their ratio is bounded above and below by positive constants. In dimensional form, expressions such as b_N^2/c_N or $\Delta_N s_N/c_N$ are multiplied by β_N^2 . Because β_N has a positive finite limit, these factors do not change any asymptotic condition.

2.3 Two exact choices of composite energy

For one relevant energy center a_N , choose

$$\mathcal{E}_N = a_N + \frac{c_N}{\beta_N}. \quad (2.7)$$

After division by $L_N(a_N)$, the weight becomes

$$W_N(e) = \begin{cases} \exp[\beta_N(e - a_N)] \left(1 - \frac{\beta_N(e - a_N)}{c_N}\right)^{c_N}, & \beta_N(e - a_N) < c_N, \\ 0, & \beta_N(e - a_N) \geq c_N. \end{cases} \quad (2.8)$$

Since $\log(1 - y) \leq -y$ for every $y < 1$,

$$0 \leq W_N(e) \leq 1 \quad (e \in \mathbb{R}). \quad (2.9)$$

This global bound, including arbitrarily low energies, is what permits a theorem without moment or tail assumptions.

For two centers $e_{-,N} < e_{+,N}$ with gap $\Delta_N = e_{+,N} - e_{-,N}$, use instead

$$\mathcal{E}_N = e_{-,N} + \frac{\Delta_N}{1 - e^{-\beta_N \Delta_N/c_N}} = e_{+,N} + \frac{\Delta_N}{e^{\beta_N \Delta_N/c_N} - 1}. \quad (2.10)$$

Then $L_N(e_{-,N}) = L_N(e_{+,N}) > 0$. Define the reference-normalized residual log-weight, obtained by subtracting its value at $e_{-,N}$, on its feasible half-line by

$$h_N(e) = \beta_N(e - e_{-,N}) + c_N \log \frac{\mathcal{E}_N - e}{\mathcal{E}_N - e_{-,N}}, \quad e < \mathcal{E}_N, \quad (2.11)$$

and set $e^{h_N(e)} = 0$ at and above the cutoff. This reference normalization is distinct from probability normalization by $\mathbb{E}_{P_N} e^{h_N(E_N)}$. It satisfies

$$h_N(e_{-,N}) = h_N(e_{+,N}) = 0, \quad h'_N(e) = \beta_N - \frac{c_N}{\mathcal{E}_N - e}, \quad h''_N(e) = -\frac{c_N}{(\mathcal{E}_N - e)^2}. \quad (2.12)$$

In particular, $h_N \geq 0$ between the centers and $h_N \leq 0$ outside them whenever feasible. Balancing the two center values is a physical choice of total energy, not an independently imposed bias. It need not balance the integrated probabilities of two broad phases.

Lemma 2.3 (Bounds for the two-center calibration). *Suppose $\Delta_N \rightarrow \infty$ and $c_N/\Delta_N \rightarrow \infty$. For all large N , with a constant C depending only on fixed bounds for β_N ,*

$$|h'_N(e_{\pm,N})| \leq C\Delta_N/c_N, \quad (2.13)$$

$$\sup_{e < \mathcal{E}_N} h_N(e) \leq C\Delta_N^2/c_N, \quad (2.14)$$

$$h_N(e) \leq C\frac{\Delta_N}{c_N}|e - e_{i,N}| \quad (i \in \{-, +\}, e < \mathcal{E}_N). \quad (2.15)$$

If $s_N = o(\Delta_N)$ and $c_N \gg \Delta_N s_N$, then for each fixed R ,

$$\sup_{|e - e_{i,N}| \leq R s_N} |h_N(e)| \longrightarrow 0, \quad (2.16)$$

provided $s_N \rightarrow \infty$. These windows are below the cutoff for all sufficiently large N .

Proof. Put $t_N = \beta_N \Delta_N / c_N \rightarrow 0$. The residual reservoir energies at the lower and upper centers are $\Delta_N / (1 - e^{-t_N})$ and $\Delta_N / (e^{t_N} - 1)$, respectively. Thus their inverse temperatures are

$$\beta_N \frac{1 - e^{-t_N}}{t_N} \quad \text{and} \quad \beta_N \frac{e^{t_N} - 1}{t_N}.$$

Their difference from β_N is $O(t_N)$, giving (2.13). On the interval between the centers, the residual reservoir energy is $(c_N / \beta_N)[1 + O(t_N)]$, so $0 \leq -h_N'' \leq C / c_N$. Interpolation between two zero endpoints gives

$$0 \leq h_N(e) \leq \frac{C}{2c_N} (e - e_{-,N})(e_{+,N} - e).$$

For completeness, subtract the right side from h_N : the difference is convex and vanishes at both endpoints, hence is nonpositive throughout the interval. Outside the interval, concavity gives $h_N \leq 0$. This proves (2.14). A concave differentiable function lies below each tangent, proving (2.15).

The cutoff is at distance asymptotic to c_N / β_N from either center. On every indicated window, $-h_N'' \leq C / c_N$ for large N . Taylor's theorem and the endpoint slope estimate give

$$|h_N(e)| \leq C \frac{\Delta_N R s_N + R^2 s_N^2}{c_N} \longrightarrow 0.$$

□

3 Sharp reservoir scales

The main distinction is between three energy locations on the same diverging scale and two narrow phase regions separated on a larger scale. The following theorem needs only a weak energy limit.

3.1 Three support points determine a quadratic threshold

Theorem 3.1 (Weak-support criterion). *Suppose that, under the canonical reference law,*

$$\frac{E_N - a_N}{b_N} \Rightarrow X, \quad b_N \rightarrow \infty, \quad (3.1)$$

and that the law of X has at least three distinct points in its support. For any prescribed exponents $c_N > 0$,

$$[\exists \text{ admissible } \mathcal{E}_N : d_{\text{TV}}(Q_N, P_N) \rightarrow 0] \iff c_N \gg b_N^2. \quad (3.2)$$

The centered choice (2.7) suffices. The necessity applies to every total-energy tuning, without a density, moment, or local-limit assumption.

We first establish the geometric inequality used for necessity.

Lemma 3.2 (Three-energy chord defect). *For $0 < \theta < 1$ and $q > 0$, define*

$$f_\theta(q) = \log[(1 - \theta)e^q + \theta] - (1 - \theta)q. \quad (3.3)$$

Then

$$f_\theta(q) \geq \frac{\theta(1 - \theta)}{2e} \min\{q^2, q\}. \quad (3.4)$$

If $g(e) = \beta_N e + c_N \log(\mathcal{E}_N - e)$ and $e_1 < e_2 < e_3 < \mathcal{E}_N$, set

$$D = e_3 - e_1, \quad \theta = \frac{e_2 - e_1}{D}, \quad q = \log \frac{\mathcal{E}_N - e_1}{\mathcal{E}_N - e_3}. \quad (3.5)$$

The exact identities are

$$g(e_3) - g(e_1) = \beta_N D - c_N q, \quad (3.6)$$

$$g(e_2) - (1 - \theta)g(e_1) - \theta g(e_3) = c_N f_\theta(q). \quad (3.7)$$

Proof. We have $f_\theta(0) = f'_\theta(0) = 0$ and

$$f''_\theta(q) = \frac{\theta(1 - \theta)e^q}{[(1 - \theta)e^q + \theta]^2}.$$

For $0 \leq q \leq 1$, the denominator is at most e^{2q} , giving $f''_\theta(q) \geq e^{-1}\theta(1 - \theta)$. Integrating twice proves (3.4) for $q \leq 1$. Convexity and $f_\theta(0) = 0$ imply that $f_\theta(q)/q$ is nondecreasing for $q > 0$, which proves it for $q \geq 1$. For the identities, the linear energy term cancels in the chord and $\mathcal{E}_N - e_2 = (1 - \theta)(\mathcal{E}_N - e_1) + \theta(\mathcal{E}_N - e_3)$. Substitution gives both formulas. $\square$

Proof of Theorem 3.1. For sufficiency, use (2.8). Weak convergence implies $E_N - a_N = O_{P_N}(b_N)$. If $c_N \gg b_N^2$, the event $|\beta_N(E_N - a_N)/c_N| \leq 1/2$ has probability tending to one. On it, Taylor's theorem gives

$$\log W_N(E_N) = -\frac{\beta_N^2(E_N - a_N)^2}{2c_N} \left[1 + O\left(\frac{|E_N - a_N|}{c_N}\right) \right] \rightarrow 0$$

in probability. The uniform bound (2.9) and Lemma 2.2 prove convergence, and also show admissibility for all large N . Changing finitely many choices of total energy has no effect on the conclusion.

For necessity, let an arbitrary admissible tuning give convergence, and write $R_N = L_N/Z_N$. Choose $\varepsilon_N \downarrow 0$ slowly enough that $P_N(|R_N - 1| > \varepsilon_N) \rightarrow 0$. On the complementary good set, $R_N > 0$ and $|\log R_N| \leq 2\varepsilon_N$ for all large N . Positivity puts every good energy strictly below the reservoir cutoff.

Choose three bounded open intervals $I_1 < I_2 < I_3$, with positive gaps, around three distinct support points of X . Each has positive limiting lower probability by the Portmanteau theorem. Therefore each event $\{(E_N - a_N)/b_N \in I_j\}$ intersects the good set for all large N . Select one corresponding energy $e_{j,N}$ in each intersection. No probability density or positive mass at an individual selected value is required: the selection is from a nonempty event. The parameters in (3.5) obey $D_N \asymp b_N$ and $\theta_N \in [\delta, 1 - \delta]$ for some fixed $\delta > 0$. The three log likelihoods are $o(1)$, so (3.6)–(3.7) imply

$$c_N q_N = \beta_N D_N + o(1), \quad c_N f_{\theta_N}(q_N) = o(1). \quad (3.8)$$

The first quantity diverges. The chord lower bound then forces $q_N \rightarrow 0$: otherwise $c_N \min(q_N^2, q_N)$ is bounded below along a subsequence by a positive multiple of $c_N q_N \rightarrow \infty$. It follows that $c_N q_N^2 \rightarrow 0$. Finally,

$$\frac{D_N^2}{c_N} = \frac{c_N q_N^2}{\beta_N^2} [1 + o(1)] \rightarrow 0.$$

Since $D_N \asymp b_N$, this is the claimed necessity. $\square$

Proposition 3.3 (The two-atom exception). *If the canonical energy law has exactly two atoms $e_- < e_+$, with any positive probabilities, then for every $c > 0$ the choice (2.10) reproduces the complete canonical state law exactly.*

Proof. The residual weight has equal positive values at both atoms, so its normalized value is one P -almost surely. Apply Lemma 2.1. $\square$

For example, a nondegenerate Gaussian energy limit on the $\sqrt{N}$ scale gives $c_N \gg N$. Three or more positive-weight phase energies in a limiting E_N/N law give $c_N \gg N^2$. A continuous non-Gaussian limit also satisfies the theorem, on whatever scale is established for that model. A two-atom *scaling limit* is different from an exactly two-atom law: fluctuations hidden by the coarse scaling can impose a larger requirement than Proposition 3.3 reveals.

3.2 Two phases with unequal energy scales

The next result states precisely how fluctuations within a phase supply the third relevant energy. Phase measures here and below are positive probability measures. They need not have disjoint supports. They can also be defined using labels on an enlarged probability space, provided E_N is the actual subsystem energy throughout.

Theorem 3.4 (Two-scale necessity). *Suppose the energy law admits the positive decomposition*

$$P_N^E = w_{-,N}P_{-,N} + w_{+,N}P_{+,N} + \delta_N R_N^{\text{exc}}, \quad w_{-,N} + w_{+,N} + \delta_N = 1, \quad (3.9)$$

where $\liminf_N w_{i,N} > 0$ for $i = -, +$. Let $\Delta_N = e_{+,N} - e_{-,N} \rightarrow \infty$ and $s_N \rightarrow \infty$ with $s_N = o(\Delta_N)$. In at least one phase, say the lower phase,

$$(E_N - e_{-,N})/s_N \Rightarrow X_- \quad \text{under } P_{-,N}, \quad (3.10)$$

where $\text{supp } X_-$ has at least two distinct points. In the other phase, assume $(E_N - e_{+,N})/\Delta_N \rightarrow 0$ in probability. Then any admissible total-energy tuning for which $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ must satisfy

$$c_N \gg \Delta_N s_N. \quad (3.11)$$

The same assertion holds with the phase roles interchanged.

Proof. The good set from the previous proof has probability tending to one under each phase law: its complementary phase probability is bounded by the unconditional complementary probability divided by $w_{i,N}$. Choose two bounded separated neighborhoods of support points of X_- . They supply good energies $e_{1,N} < e_{2,N}$ with $e_{2,N} - e_{1,N} \asymp s_N$ and $e_{j,N} = e_{-,N} + O(s_N)$. Concentration in the upper phase supplies a good energy $e_{3,N} = e_{+,N} + o(\Delta_N)$; to make this selection explicit, choose a deterministic sequence $r_N \downarrow 0$ slowly enough that $P_{+,N}(|E_N - e_{+,N}| \leq r_N \Delta_N) \rightarrow 1$. Thus $D_N \asymp \Delta_N$ and $\theta_N(1 - \theta_N) \asymp s_N/\Delta_N$.

The identities (3.8) still hold. Since β_N is bounded above and away from zero and $D_N \rightarrow \infty$, they imply $q_N \asymp \Delta_N/c_N$. The chord bound is therefore at least a positive constant times

$$c_N \frac{s_N}{\Delta_N} \min(q_N^2, q_N) \asymp \min\left\{\frac{\Delta_N s_N}{c_N}, s_N\right\}.$$

It must tend to zero. Since $s_N \rightarrow \infty$, this forces $\Delta_N s_N/c_N \rightarrow 0$. If the upper phase provides the two points, the same argument has $1 - \theta_N \asymp s_N/\Delta_N$. $\square$

This obstruction is local to positive-probability phase neighborhoods. It requires neither a Gaussian density nor control of the interphase valley. In particular, extensive separation and nondegenerate weak $\sqrt{N}$ fluctuations imply the necessary scale $c_N \gg N^{3/2}$. The converse needs more information: under two-center calibration the reservoir can amplify rare energies between the peaks by as much as $\exp[O(\Delta_N^2/c_N)]$.

3.3 A positive-phase sufficient criterion

Theorem 3.5 (Positive decomposition and exceptional mass). *Assume the positive decomposition (3.9), with $\Delta_N \rightarrow \infty$, $s_N \rightarrow \infty$, and $s_N = o(\Delta_N)$. Suppose that, for some fixed $\eta > 0$, $M < \infty$, and N_0 ,*

$$\sup_{N \geq N_0, i \in \{-, +\}} \mathbb{E}_{P_{i,N}} \exp\left(\eta \frac{|E_N - e_{i,N}|}{s_N}\right) \leq M. \quad (3.12)$$

There is a fixed constant C depending only on bounds for β_N such that the conditions

$$c_N \gg \Delta_N s_N, \quad \delta_N \exp(C\Delta_N^2/c_N) \rightarrow 0 \quad (3.13)$$

ensure $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ with the two-center calibration (2.10). Positive limiting phase weights and local limit theorems are not required for this sufficient statement.

Proof. Let $W_N = e^{h_N}$, with value zero at and above the cutoff. Since $c_N/\Delta_N \rightarrow \infty$, Lemma 2.3 applies. The exponential moment implies tightness on the phase scale. Together with (2.16), it gives $W_N \rightarrow 1$ in each phase probability. The tangent bound gives globally

$$W_N(e) \leq \exp\left(\epsilon_N \frac{|e - e_{i,N}|}{s_N}\right), \quad \epsilon_N = C\Delta_N s_N / c_N \rightarrow 0.$$

For large N , $2\epsilon_N \leq \eta$, so $\mathbb{E}_{P_{i,N}} W_N^2 \leq M$. This is uniform integrability, and therefore $\mathbb{E}_{P_{i,N}} |W_N - 1| \rightarrow 0$ for each phase. On the exceptional component,

$$\delta_N \mathbb{E}_{R_N^{\text{exc}}} |W_N - 1| \leq \delta_N \{1 + \exp(C\Delta_N^2/c_N)\} \rightarrow 0.$$

Adding the positive components and using Lemma 2.2 completes the proof. $\square$

Corollary 3.6 (A sharp two-phase criterion). *Assume the positive decomposition, scale hypotheses, and moment bound of Theorem 3.5. Suppose additionally that the phase weights have positive lower limits, at least one scaled phase law has a weak limit with two distinct support points, and*

$$\delta_N \leq \exp(-b\Delta_N/s_N) \quad \text{for some fixed } b > 0. \quad (3.14)$$

Then a total-energy tuning with $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ exists if and only if $c_N \gg \Delta_N s_N$. In particular, if $\Delta_N \asymp N$ and $s_N \asymp \sqrt{N}$, the sharp scale is $c_N \gg N^{3/2}$.

Proof. The moment bound makes the other phase tight on scale s_N , hence concentrated on the smaller-than-gap scale required by Theorem 3.4. For sufficiency, $\Delta_N^2/c_N = o(\Delta_N/s_N)$ and $\Delta_N/s_N \rightarrow \infty$, so (3.14) implies the second condition in (3.13). $\square$

More generally, if $\delta_N \leq e^{-t_N}$ with $t_N \rightarrow \infty$, the sufficient condition

$$c_N \gg \max\{\Delta_N s_N, \Delta_N^2/t_N\} \quad (3.15)$$

follows immediately. It is not a necessary condition expressed solely in terms of the exceptional mass: an exceptional component near the phase endpoints may receive much less amplification than the global maximum used in the proof. Nor can a signed remainder in a partition function expansion be substituted for the positive measure $\delta_N R_N^{\text{exc}}$. The exponential moment must concern the energy actually exchanged with the reservoir.

3.4 An alternative criterion allowing interfacial tails

The positive-phase exponential moment is one route to sufficiency. A uniform bound on the full energy density gives another. We state the bound explicitly, rather than deducing it from a local Gaussian limit. Let p_N be the density of P_N^E with respect to Lebesgue measure and let φ_v denote the centered Gaussian density of variance v . Assume

$$\Delta_N/N \longrightarrow \ell > 0, \quad (3.16)$$

$$\sqrt{N} p_N(e_{i,N} + \sqrt{N}z) \longrightarrow w_i \varphi_{v_i}(z) \quad \text{in } L^1([-R, R]) \quad \text{for every } R < \infty, \quad (3.17)$$

where $w_i > 0$, $w_- + w_+ = 1$, and $v_i > 0$. On the interval between the centers, set $x_N(e) = \min(e - e_{-,N}, e_{+,N} - e)$. Suppose fixed $A, a > 0$ and $\alpha \in [1/2, 1]$ give

$$p_N(e) \leq \frac{A}{\sqrt{N}} \exp \left[-a \min \left\{ \frac{x_N(e)^2}{N}, x_N(e)^\alpha \right\} \right], \quad e \in [e_{-,N}, e_{+,N}]. \quad (3.18)$$

Constants carry the necessary fixed energy units. The minimum in this bound permits a transition from a Gaussian cost to a subextensive interfacial cost. No envelope is imposed outside the phase interval.

It is useful to give the sufficient argument for a general concave residual weight; the exact physical reservoir is then a direct application. Let I_N be an interval of feasible energies, with h_N concave and twice continuously differentiable in its interior, and set $e^{h_N} = 0$ outside I_N . Require that I_N contain the two centers and all their fixed $R\sqrt{N}$ neighborhoods for large N . Suppose

$$h_N(e_{-,N}) = h_N(e_{+,N}) = 0, \quad 0 \leq -h_N''(e) \leq K\kappa_N \quad (3.19)$$

between the centers and on each such neighborhood, with a constant K independent of N and R .

Proposition 3.7 (Sufficiency with an interior density envelope). *Under (3.16)–(3.19), if $\kappa_N N^{3/2} \rightarrow 0$, then*

$$\mathbb{E}_{P_N} |e^{h_N(E_N)} - 1| \longrightarrow 0. \quad (3.20)$$

The normalized reweighted law therefore converges in total variation.

Proof. Concavity and the endpoint values imply that the endpoint derivatives are bounded in absolute value by $K\kappa_N \Delta_N$; for example, integrate h_N' between the zero endpoints and use its $K\kappa_N$ Lipschitz bound. Hence, uniformly for $|z| \leq R$,

$$|h_N(e_{i,N} + \sqrt{N}z)| \leq K\kappa_N \Delta_N \sqrt{N}R + \frac{1}{2}K\kappa_N N R^2 \longrightarrow 0. \quad (3.21)$$

Interpolation between the endpoints also gives, inside the interval,

$$0 \leq h_N(e) \leq \frac{1}{2}K\kappa_N (e - e_{-,N})(e_{+,N} - e) \leq C\kappa_N N x_N(e). \quad (3.22)$$

Set $\lambda_N = \kappa_N N^{3/2}$. For $R\sqrt{N} \leq x \leq \Delta_N/2$, the ratios of the last bound to the two costs in (3.18) satisfy

$$\frac{C\kappa_N N x}{x^2/N} \leq C\lambda_N/R, \quad \frac{C\kappa_N N x}{x^\alpha} \leq C\lambda_N N^{1/2-\alpha}.$$

For each fixed $R > 0$, both tend to zero. The reservoir-weighted interior mass outside the two phase windows is consequently bounded for large N by

$$\frac{2A}{\sqrt{N}} \int_{R\sqrt{N}}^{\Delta_N/2} \exp \left[-\frac{a}{2} \min \left\{ x^2/N, x^\alpha \right\} \right] dx. \quad (3.23)$$

Since $e^{-a \min(u,v)/2} \leq e^{-au/2} + e^{-av/2}$, the Gaussian part has limiting upper bound $2A \int_R^\infty e^{-az^2/2} dz$, and the other part tends to zero for fixed $R > 0$ (its integral over the positive half-line is finite, and there is an additional $N^{-1/2}$ factor).

Outside the phase interval, concavity gives $h_N \leq 0$ wherever feasible, and the weight is zero elsewhere. These tails cannot be amplified. Equation (3.17), with the weights summing to one, implies

$$\lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} P_N \left(E_N \notin \bigcup_i [e_{i,N} - R\sqrt{N}, e_{i,N} + R\sqrt{N}] \right) = 0.$$

On each fixed phase window (3.21) makes the weight uniformly approach one. On the remaining interior use (3.23) and the corresponding unweighted tail; on the exterior use $|e^{h_N} - 1| \leq 1$. Taking $N \rightarrow \infty$ first and then $R \rightarrow \infty$ proves (3.20). $\square$

Corollary 3.8 (Physical two-phase threshold with interfacial tails). *Under (3.16)–(3.18), an exact reservoir (2.2) can reproduce the canonical law in total variation by some total-energy tuning if and only if $c_N \gg N^{3/2}$.*

Proof. For sufficiency choose (2.10). If $c_N \gg N^{3/2}$, the residual reservoir energy is $(c_N/\beta_N)[1 + o(1)]$ uniformly between the centers and in each fixed phase neighborhood. Thus (3.19) holds with $\kappa_N = 1/c_N$, and Proposition 3.7 applies.

For necessity, divide the energy line at the midpoint of the centers. The local limits and their total weight one imply that these two positive conditional phase measures have weights tending to w_i and scaled energy laws converging weakly to $\mathcal{N}(0, v_i)$. Indeed their mass in every fixed scaled compact interval has the stated limit, and the limiting Gaussian mass tends to the entire phase weight as the interval increases; no mass can escape on either side of the midpoint. Theorem 3.4 with $s_N = \sqrt{N}$ then gives the required condition for any tuning. $\square$

The density hypotheses are stronger than local coexistence limits. They are assumptions to be verified for the specified microscopic Hamiltonian and its energy, not consequences of identifying two peaks. Proposition 3.7 concerns continuous densities; Theorems 3.1, 3.4, and 3.5 apply without change to discrete, continuous, or mixed laws.

4 Microscopic realizations

We now verify the probabilistic hypotheses for two microscopic Hamiltonians. The mean-field example permits exact finite-system calculations. The short-range example requires positive contour measures and a transfer estimate between bond and spin energies. In both models the reservoir threshold is a consequence of their established phase structure, rather than a new claim about the location or order of their transitions.

4.1 The three-state Curie–Weiss–Potts model

Let $n_1 + n_2 + n_3 = N$ be the occupation numbers of N three-state spins and take

$$U_N = -\frac{J}{2N} \sum_{j=1}^3 n_j^2, \quad J > 0, \quad \beta J = b := 4 \log 2. \quad (4.1)$$

Optionally add $2aN$ independent quadratic momentum coordinates, with $a > 0$ fixed and $2aN$ integral on the size sequence under consideration. Their canonical energy K_N has the Gamma

distribution of shape aN and rate β . Set $E_N = U_N + K_N$. We also allow $a = 0$, meaning no momenta and $K_N = 0$.

Theorem 4.1 (Mean-field microscopic reservoir threshold). *For every fixed $J > 0$ and $a \geq 0$ in the model just defined, an admissible total-energy tuning for which $d_{\text{TV}}(Q_N, P_N) \rightarrow 0$ exists if and only if $c_N \gg N^{3/2}$. The two-center calibration uses*

$$e_{-,N} = N \left(-\frac{J}{4} + \frac{a}{\beta} \right), \quad e_{+,N} = N \left(-\frac{J}{6} + \frac{a}{\beta} \right). \quad (4.2)$$

We give the phase analysis and uniform bounds needed for this statement. The known mean-field phase structure and fluctuation theory are treated more generally in [12, 19]. The calculation here makes the normalization and the degenerate disordered spin-energy fluctuation explicit.

Write $p_j = n_j/N$ and use (p_1, p_2) as coordinates on the probability simplex, with $p_3 = 1 - p_1 - p_2$. The rate potential, up to an additive constant, is

$$I(p) = \sum_{j=1}^3 p_j \log p_j - \frac{b}{2} \sum_{j=1}^3 p_j^2. \quad (4.3)$$

Its minima are

$$p^{(d)} = (1/3, 1/3, 1/3), \quad p^{(o)} \in \{(2/3, 1/6, 1/6), (1/6, 2/3, 1/6), (1/6, 1/6, 2/3)\}. \quad (4.4)$$

There are three ordered minima and one disordered minimum. Here is a direct completeness check. A boundary point cannot minimize I , because transferring a small positive mass to a zero coordinate lowers the entropy term by order $\varepsilon |\log \varepsilon|$ and changes the quadratic term by only $O(\varepsilon)$. At an interior stationary point, all $\log p_j - bp_j$ are equal. Strict concavity of $\log x - bx$ permits at most two coordinate values. If two coordinates share the larger value x and the remaining value is $y < x$, stationarity gives $b = \log(x/y)/(x - y) > 1/x$. The second variation exchanging the two large coordinates is then negative. A nonsymmetric minimum must therefore have one large coordinate r and two equal small coordinates $(1 - r)/2$, with $r > 1/3$. Its stationary equation is

$$\log \frac{2r}{1-r} - \frac{b}{2}(3r - 1) = 0. \quad (4.5)$$

The derivative of the left side is $1/[r(1-r)] - 3b/2$, which has at most two zeros. Rolle's theorem therefore allows at most three roots of (4.5); the three roots are $1/3, 1/2, 2/3$. At $r = 1/2$ the second variation along this curve is $4 - 6 \log 2 < 0$. The other candidates have the same value $I_{\min} = -\log 3 - (2/3) \log 2$. Their Hessians are positive definite:

$$H_d = (3 - b) \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \det H_d = 3(3 - b)^2, \quad (4.6)$$

$$H_o = \begin{pmatrix} 15/2 - 2b & 6 - b \\ 6 - b & 12 - 2b \end{pmatrix}, \quad \det H_o = 3(3 - b)(6 - b). \quad (4.7)$$

This proves the complete minimum classification used below.

Partition the simplex into the four Voronoi cells of these minima, resolving ties deterministically. All four cells have positive limiting probabilities. Uniform Stirling expansion on compact interior sets gives the unnormalized occupation weight

$$\frac{N!}{\prod_j n_j!} e^{-\beta U_N(n)} = \frac{1 + o(1)}{2\pi N \sqrt{p_1 p_2 p_3}} e^{-NI(p)}. \quad (4.8)$$

In a fixed $N^{-1/2}$ neighborhood of a minimum, the two-dimensional lattice has mesh $N^{-1/2}$ after rescaling and hence density N . Taylor expansion of I in (4.8) and Gaussian Riemann sums give a phase contribution proportional to $(p_1 p_2 p_3)^{-1/2} (\det H_i)^{-1/2}$ and a conditional occupation limit $\mathcal{N}(0, H_i^{-1})$. The uniform bound proved below justifies discarding the tails of these Riemann sums. Thus, with minus denoting the combined ordered phases,

$$r_* := \sqrt{\frac{2(3-b)}{6-b}}, \quad w_- := \frac{3r_*}{1+3r_*}, \quad w_+ := \frac{1}{1+3r_*}. \quad (4.9)$$

The tangent gradient of $u(p) = -J \sum_j p_j^2/2$ is $(-J/2, 0)$ at the displayed ordered minimum and zero at the disordered minimum. The delta method and independence of the kinetic energy give the conditional weak energy limits

$$\frac{E_N - e_{i,N}}{\sqrt{N}} \Rightarrow \mathcal{N}(0, v_i), \quad v_- := \frac{a}{\beta^2} + \frac{J^2}{6(3-b)}, \quad v_+ := \frac{a}{\beta^2}. \quad (4.10)$$

For $a = 0$, $\mathcal{N}(0, 0)$ means the point mass at zero. Indeed the disordered potential energy fluctuates only on scale one, because its first derivative on the simplex vanishes. The ordered phase has a strictly positive $\sqrt{N}$ variance even without momenta.

Lemma 4.2 (Uniform positive-phase energy moments). *For each fixed $a \geq 0$, there are $\eta > 0$ and $M < \infty$ such that the two positive energy phase laws defined by the Voronoi partition satisfy*

$$\sup_{N \geq N_0, i \in \{-, +\}} \mathbb{E} \left[\exp \left(\eta \frac{|E_N - e_{i,N}|}{\sqrt{N}} \right) \middle| i \right] \leq M. \quad (4.11)$$

Here N_0 is fixed and large enough that every conditioning phase has positive probability. Only these sufficiently large sizes are needed for the asymptotic theorem.

Proof. Let $\mathcal{M}$ be the four minima. Compactness, the complete minimum classification, and their positive Hessians give $I(p) - I_{\min} \geq c_0 \text{dist}(p, \mathcal{M})^2$ globally. The Gaussian neighborhoods in (4.8) give a partition function lower bound $ce^{-NI_{\min}}$. On a fixed interior subset, Stirling bounds then give

$$P_N(n) \leq \frac{C}{N} e^{-cN \text{dist}(n/N, \mathcal{M})^2}.$$

The same bound holds globally with smaller c . To verify the boundary, choose a boundary strip disjoint from $\mathcal{M}$, where the rate gap is at least $\zeta > 0$. The multinomial type bound $N! / \prod n_j! \leq e^{-N \sum p_j \log p_j}$ gives $P_N(n) \leq C e^{-N[I(p) - I_{\min}]}$ there. Splitting the exponent between the gap and distance bounds recovers the missing $1/N$, since $N e^{-N\zeta/2}$ is bounded.

In the cell of $p^{(i)}$, the distance to $\mathcal{M}$ is $|p - p^{(i)}|$ and the Lipschitz property of u gives $|U_N - Nu(p^{(i)})|/\sqrt{N} \leq C\sqrt{N}|p - p^{(i)}|$. Completing the square in the resulting lattice Gaussian sum shows

$$\frac{C}{N} \sum_{n_1, n_2} e^{-cN|n/N - p^{(i)}|^2 + \eta C\sqrt{N}|n/N - p^{(i)}|} \leq C_\eta.$$

The phase probabilities are bounded below for large N , so conditioning preserves the bound. For $a > 0$, the centered Gamma transform is

$$\mathbb{E} e^{t(K_N - aN/\beta)/\sqrt{N}} = e^{-ta\sqrt{N}/\beta} \left(1 - \frac{t}{\beta\sqrt{N}} \right)^{-aN}.$$

It is uniformly bounded for fixed t and all sufficiently large N . Using $e^{t|x|} \leq e^{tx} + e^{-tx}$ and independence completes the proof. $\square$

Proof of Theorem 4.1. The gap of the two energy centers is $NJ/12$. Lemma 4.2 supplies a positive decomposition with no exceptional component and the required $\sqrt{N}$ moment bounds. Equation (4.10) supplies a nondegenerate weak limit in the ordered phase. Apply Corollary 3.6. $\square$

The kinetic variables therefore have a useful but limited role: they make both energy peaks continuous with positive Gaussian variance and permit the exact integral formulas below. They are not required for the reservoir exponent itself.

4.2 Continuous energy limits and exact finite-system formulas

For $a > 0$, the standardized Gamma density converges uniformly to the Gaussian density of variance a/β^2 . One way to check uniformity is to apply Stirling's formula on compact standardized intervals and then use the Gamma density's unimodality: outside such an interval its supremum is attained at the nearest endpoint once the mode is inside. Increasing the interval bounds both limiting tails uniformly. Convolution with the conditional spin-energy weak limits (the $a = 0$ case of (4.10)) therefore gives the local L^1 density limits (3.17) with weights (4.9). There is also the stronger interior bound

$$p_N(E) \leq \frac{C}{\sqrt{N}} \sum_{i \in \{-, +\}} \exp \left[-c \frac{(E - e_{i,N})^2}{N} \right] \quad (e_{-,N} \leq E \leq e_{+,N}). \quad (4.12)$$

To see this without a Gaussian-mixture assumption, write g_N for the kinetic density. The allowed kinetic energies in this interval obey $0 < K \leq C_0 N$. Its logarithmic curvature is $-(aN - 1)/K^2 \leq -c_1/N$ for all large N , and its maximum is $O(N^{-1/2})$. Expanding at its mode, which differs from aN/β by a constant, gives $g_N(K) \leq CN^{-1/2} e^{-c(K - aN/\beta)^2/N}$ on this whole interval. For a nearest minimum $p^{(i)}$, the Lipschitz property of u yields

$$|p - p^{(i)}|^2 + |E/N - u(p) - a/\beta|^2 \geq c\{|p - p^{(i)}|^2 + |E/N - u(p^{(i)}) - a/\beta|^2\}.$$

Multiply the occupation bound from Lemma 4.2 by the Gamma bound, and sum over each cell. The lattice Gaussian sum is $O(N)$ and cancels the occupation prefactor N^{-1} , proving (4.12). This also controls contamination from the other phase in the local density limit.

For finite-system evaluation set $A = aN > 0$ and $m(n) = N!/(n_1!n_2!n_3!)$. The canonical occupation law and its physical reservoir counterpart are exactly

$$P_N(n) = \frac{m(n)e^{-\beta U_N(n)}}{\sum_{n'} m(n')e^{-\beta U_N(n')}}, \quad (4.13)$$

$$Q_N(n) = \frac{m(n)[\mathcal{E}_N - U_N(n)]_+^{A+c_N}}{\sum_{n'} m(n')[\mathcal{E}_N - U_N(n')]_+^{A+c_N}}. \quad (4.14)$$

Conditional on n , K_N is canonical Gamma(A, β), whereas

$$\frac{K_N}{\mathcal{E}_N - U_N(n)} \Bigg| n \sim \text{Beta}(A, c_N + 1) \quad \text{under } Q_N. \quad (4.15)$$

Indeed integration of the product of subsystem and reservoir densities gives

$$\int_0^{\mathcal{E}_N - U_N(n)} K^{A-1} [\mathcal{E}_N - U_N(n) - K]^{c_N} dK = B(A, c_N + 1) [\mathcal{E}_N - U_N(n)]^{A+c_N}.$$

The exponent is $A + c_N$, and the second Beta parameter is $c_N + 1$. For $a = 0$, the occupation formulas remain valid with $A = 0$, but the conditional energy is the point mass $K_N = 0$; no shape-zero Gamma or Beta distribution is used.

Let G_A be the CDF of a unit-rate Gamma law and $I_x(A, c_N + 1)$ the regularized incomplete Beta function, extended by zero for $x \leq 0$ and one for $x \geq 1$. Then, for $a > 0$,

$$P_N(E_N \leq t) = \sum_n P_N(n) G_A(\beta[t - U_N(n)]), \quad (4.16)$$

$$Q_N(E_N \leq t) = \sum_{n: Q_N(n) > 0} Q_N(n) I_{[t - U_N(n)]/[\varepsilon_N - U_N(n)]}(A, c_N + 1), \quad (4.17)$$

where $G_A(x) = 0$ for $x \leq 0$. These are finite sums, with no histogram approximation. The log likelihood in (2.3) is strictly concave below its cutoff and tends to $-\infty$ at both ends of that half-line. For $a > 0$ it is not constant on the canonical support, so its normalized maximum exceeds zero. Its positive set is an interval (t_L, t_R) , determined by two scalar roots. Consequently

$$d_{\text{TV}}(Q_N, P_N) = Q_N(t_L < E_N < t_R) - P_N(t_L < E_N < t_R). \quad (4.18)$$

Equations (4.16)–(4.17) evaluate this expression directly. An interval endpoint may lie outside the subsystem support; the CDF formula already handles that case. For the pure-spin model one instead sums the positive likelihood differences over the discrete occupation law.

4.3 A short-range Potts theorem

Fix a dimension $d \geq 2$ and an integer $q \geq q_0(d)$, independent of system size. The threshold $q_0(d)$ is fixed once at the start of Appendix A: it exceeds $2d$, and it is large enough for the cited contour results and for one additional exponential slack in their cluster-expansion criterion. On the periodic nearest-neighbor torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^d$, let $N = L^d$ and

$$U_N(\sigma) = - \sum_{\{x,y\} \in \mathcal{B}_L} \mathbf{1}_{\{\sigma_x = \sigma_y\}}, \quad \sigma \in \{1, \dots, q\}^{\Lambda_L}. \quad (4.19)$$

There are dN bonds. The unit coupling fixes the energy units. The coexistence inverse temperature is denoted β_c .

Theorem 4.3 (Short-range microscopic reservoir threshold). *For every fixed $d \geq 2$ and fixed $q \geq q_0(d)$, the canonical law of (4.19) at β_c can be reproduced in total variation by an exact reservoir (2.2), with some total-energy tuning, if and only if $c_N \gg N^{3/2}$. The sufficient choice is (2.10) with centers Nu_o and Nu_d , where $u_o < u_d$ are the two coexistence energy densities.*

In particular, this statement applies to the plain two-dimensional spin model, with no auxiliary kinetic energy. The proof uses two different positive phase partitions. Midpoint energy events provide the weak fluctuations needed for necessity. Contour phase events in the Edwards–Sokal joint space provide the uniform moments needed for sufficiency. There is no requirement that these partitions coincide.

The detailed proof is in Appendix A. The published finite-volume free energies provide weak Gaussian energy limits with positive phase weights. A uniform conditional-variance bound proves that each stable-phase variance is positive. These inputs give necessity through Theorem 3.4. For sufficiency, the logarithms of positive restricted contour partition functions have second derivatives bounded in absolute value by CN in the stated finite-size temperature window. Their moment bounds transfer from occupied-bond count to the exchanged spin energy using the conditional

binomial law of the Edwards–Sokal coupling. The exceptional contour mass decays on the surface scale, which dominates the reservoir gain when $c_N \gg N^{3/2}$. The distinction between bond count and physical energy is essential; differentiating a bond-conditioned spin law would not justify this step.

5 Canonical subsystem marginals under a shared reservoir

Two identical systems sharing a reservoir can occupy opposite phases while each separately has its canonical distribution. Their missing independence has an exact information-theoretic limit. Thermally coupled systems with negative specific heat were previously shown to develop different subsystem phases and to exchange their assignments in [29]. The results below quantify the reservoir-size conditions for this behavior in the full microscopic law and distinguish the accuracy of separate marginals from joint canonical accuracy.

5.1 Opposite-phase selection by the exact reservoir

Let P_N be a one-copy canonical law at the specified inverse temperature $\beta_N \rightarrow \beta > 0$. Assume two disjoint measurable phase events $A_{-,N}, A_{+,N}$ partition its state space, with

$$w_{i,N} := P_N(A_{i,N}) \longrightarrow w_i > 0, \quad w_- + w_+ = 1. \quad (5.1)$$

Let $e_{-,N} < e_{+,N}$ be deterministic energy centers, with $\Delta_N = e_{+,N} - e_{-,N}$, chosen so that

$$\Delta_N/N \longrightarrow \ell > 0, \quad (E_N - e_{i,N})/\sqrt{N} \mid A_{i,N} \text{ is tight.} \quad (5.2)$$

Any sequences with these two properties may serve as centers; the results below use nothing else about them. The assumptions permit discrete phase laws and a degenerate conditional energy limit. In this section N denotes the size of one copy.

Take two independent copies under $\Pi_N = P_N \otimes P_N$ and write X_1, X_2 for their complete microscopic states and S_1, S_2 for the phase labels. Their combined energy is $E_N(X_1) + E_N(X_2)$. There is no direct interaction between the copies. Couple their sum to one exact reservoir and choose

$$M_N = e_{-,N} + e_{+,N}, \quad \mathcal{E}_N = M_N + c_N/\beta_N. \quad (5.3)$$

The centered weight from (2.8) is

$$W_N = \begin{cases} \exp[\beta_N T_N](1 - \beta_N T_N/c_N)^{c_N}, & \beta_N T_N < c_N, \\ 0, & \beta_N T_N \geq c_N, \end{cases} \quad T_N = E_N(X_1) + E_N(X_2) - M_N. \quad (5.4)$$

Let $z_N = \mathbb{E}_{\Pi_N} W_N$ and $dQ_N/d\Pi_N = W_N/z_N$. The bound $0 \leq W_N \leq 1$ holds globally, including all interphase and far-tail energies.

Theorem 5.1 (Opposite phases and full marginal accuracy). *Under (5.1)–(5.2), suppose*

$$N \ll c_N \ll N^2. \quad (5.5)$$

For $O_N = \{S_1 \neq S_2\}$, the centered shared reservoir satisfies

$$d_{\text{TV}}(Q_N, \Pi_N(\cdot \mid O_N)) \longrightarrow 0. \quad (5.6)$$

Consequently, with $Q_N^{(j)}$ denoting either one-copy marginal,

$$d_{\text{TV}}(Q_N, \Pi_N) \longrightarrow w_-^2 + w_+^2, \quad (5.7)$$

$$d_{\text{TV}}(Q_N^{(j)}, P_N) \longrightarrow |w_- - 1/2|, \quad j = 1, 2. \quad (5.8)$$

Proof. On either opposite-phase assignment $T_N = O_{\Pi_N}(\sqrt{N})$ conditionally. The Taylor expansion $x + \log(1 - x) = -x^2/2 + O(|x|^3)$, together with $N/c_N \rightarrow 0$, gives $\log W_N \rightarrow 0$ in its conditional probability; the cutoff lies beyond every fixed fluctuation window. On the same-phase assignments, $T_N = \pm \Delta_N + O_{\Pi_N}(\sqrt{N})$. Because $c_N/N \rightarrow \infty$, the Taylor expansion still applies in conditional probability, but now

$$\log W_N = -\frac{\beta_N^2 \Delta_N^2}{2c_N} [1 + o_{\Pi_N}(1)] \rightarrow -\infty.$$

Thus $W_N - \mathbf{1}_{O_N} \rightarrow 0$ in probability and, by boundedness, in $L^1(\Pi_N)$. In particular, $z_N \rightarrow p_O := 2w_-w_+ > 0$. Normalizing the two nonnegative weights proves (5.6).

Conditioning a probability measure on an event of probability p changes it by $1 - p$ in total variation. This gives (5.7). The exact marginal of $\Pi_N(\cdot \mid O_N)$ is

$$\frac{1}{2}P_N(\cdot \mid A_{-,N}) + \frac{1}{2}P_N(\cdot \mid A_{+,N}).$$

The two phase supports are disjoint, so its distance from P_N is $|w_{-,N} - 1/2|$. Total variation contracts under projection, proving (5.8). $\square$

When $w_- = w_+ = 1/2$, both *complete* subsystem marginals converge to their canonical laws, while the joint distance tends to $1/2$. The result does not depend on a density approximation or a bound on the probability of a particular interfacial configuration. Tightness controls the atypical canonical mass, and the chosen reservoir cannot amplify that mass. The second copy accommodates a compensating latent-energy change, while the ordinary reservoir accommodates the smaller phase fluctuations.

5.2 The full microscopic mutual information

For $Q \ll P$, define $D_{\text{KL}}(Q \parallel P) = \int \log(dQ/dP) dQ$, with $0 \log 0 = 0$. All logarithms are natural, so information is initially in nats. For a joint law, its mutual information is $I_Q(X_1; X_2) = D_{\text{KL}}(Q \parallel Q^{(1)} \otimes Q^{(2)})$. Total variation by itself does not control this quantity on growing or continuous state spaces. We therefore prove the entropy limit separately from Theorem 5.1.

Proposition 5.2 (One bit in the full state law). *Under the hypotheses of Theorem 5.1,*

$$I_{Q_N}(X_1; X_2) \rightarrow \log 2. \quad (5.9)$$

This limit is independent of the original positive phase weights.

Proof. The function $x \log x$ is continuous and bounded on $[0, 1]$. The proof of Theorem 5.1 therefore gives $\mathbb{E}_{\Pi_N} W_N \log W_N \rightarrow 0$. Hence

$$D_{\text{KL}}(Q_N \parallel \Pi_N) = z_N^{-1} \mathbb{E}_{\Pi_N} W_N \log W_N - \log z_N \rightarrow -\log(2w_-w_+). \quad (5.10)$$

Each marginal likelihood $r_{j,N} = dQ_N^{(j)}/dP_N$ is at most $1/z_N$, and so is bounded uniformly for all large N . By (5.6) it converges in $L^1(P_N)$ to the finite- N conditional marginal ratio

$$r_N^O = \frac{\mathbf{1}_{A_{-,N}}}{2w_{-,N}} + \frac{\mathbf{1}_{A_{+,N}}}{2w_{+,N}}.$$

Both ratios lie in a common compact interval of $[0, \infty)$. Uniform continuity of $r \log r$ on that interval and convergence in probability imply convergence of its integrals. Consequently

$$D_{\text{KL}}(Q_N^{(j)} \parallel P_N) \rightarrow \frac{1}{2} \log \frac{1}{2w_-} + \frac{1}{2} \log \frac{1}{2w_+}. \quad (5.11)$$

All joint and marginal divergences are finite, because their likelihood ratios are bounded. The relative-entropy chain rule gives

$$I_{Q_N}(X_1; X_2) = D_{\text{KL}}(Q_N \| \Pi_N) - \sum_{j=1}^2 D_{\text{KL}}(Q_N^{(j)} \| P_N). \quad (5.12)$$

Substituting (5.10) and (5.11) proves (5.9). No microscopic differential entropy is used. $\square$

The limiting phase labels are uniform and perfectly anticorrelated. Equation (5.9) establishes that no additional nonzero limiting amount of mutual information remains in their internal microscopic fluctuations in the window (5.5).

5.3 Joint canonical accuracy requires a quadratic-size reservoir

Corollary 5.3 (Optimized joint threshold). *Suppose only that, for some centers a_N , $(E_N - a_N)/N$ converges weakly to a law with exactly two distinct positive-weight atoms. For any fixed number of independent identical copies $m \geq 2$, a shared physical reservoir can reproduce their complete canonical product law in total variation, by some total-energy tuning, if and only if $c_N \gg N^2$.*

Proof. After a common centering, the total energy divided by N converges to the m -fold convolution of the two-atom law. It has $m + 1 \geq 3$ distinct support points, all of positive weight. Apply Theorem 3.1 with $b_N = N$ to the combined subsystem. The number of copies is fixed, so the choice of N rather than mN does not change the asymptotic condition. $\square$

No phase central limit theorem is required for this corollary. For the two-copy assumptions above, the centered total energy $[E_N(X_1) + E_N(X_2) - M_N]/N$ has limiting atoms $-\ell, 0, \ell$. No individual limit for $e_{i,N}/N$ is needed. For the microscopic models in Section 4, an isolated copy has the sharp threshold $c_N \gg N^{3/2}$, whereas a fixed collection of at least two copies has the stronger joint threshold $c_N \gg N^2$. With balanced phase weights, a choice such as $c_N = N^{5/4}$ nevertheless gives canonical one-copy marginals under the shared bath. The isolated copy and the shared pair are different composite systems; the phase anticorrelation records the role of the additional energy store. Two extensions are given in Appendix D: selection of a fixed number of plus-phase copies among m copies, and the correlated within-phase fluctuations that appear when the capacity is only proportional to N .

5.4 Balancing the reference phase weights

The limiting phase weights at a fixed coexistence temperature need not be equal. To realize the canonical-marginal case in a microscopic model, the canonical reference itself can be chosen at a specified nearby temperature. The following elementary statement makes the change of reference explicit.

Lemma 5.4 (A bounded-energy temperature shift). *Let the spin energy U_N/N be uniformly bounded and suppose the phase conditional laws at β_c concentrate at $u_- < u_+$, with positive limiting weights w_- , w_+ . For $\beta_N = \beta_c + \delta/N + o(N^{-1})$, the new limiting weight ratio is*

$$\frac{w_-^{(\delta)}}{w_+^{(\delta)}} = \frac{w_-}{w_+} \exp[\delta(u_+ - u_-)]. \quad (5.13)$$

The new spin law conditional on either phase approaches the old phase-conditional law in total variation. Hence any conditional weak fluctuation limit, with the same center and scale, persists.

Proof. The change of canonical measure has an unnormalized likelihood $\exp[-(\beta_N - \beta_c)U_N]$. It is bounded above and below by positive constants uniformly in N . On phase i it converges in probability to $e^{-\delta u_i}$. Bounded convergence gives the limiting phase normalizations and (5.13). After normalizing within either phase, the likelihood converges to one in L^1 , proving the conditional total-variation assertion. $\square$

Writing $\ell = u_+ - u_-$, asymptotic balance is obtained by

$$\beta_N = \beta_c - \frac{\log(w_-/w_+)}{N\ell} + o(N^{-1}). \quad (5.14)$$

The negative sign is consistent with warming a system whose ordered, lower-energy phase initially has excess weight. For the short-range model, $w_-/w_+ = q$, and the required shift is $-\log q/(N\ell)$. For the three-state mean-field model, the ratio is $3r_*$ from (4.9) and $\ell = J/12$. Both spin Hamiltonians are bounded per particle, so Lemma 5.4 applies directly. For the short-range positive contour decomposition, retain the conditional bond kernel $K_{\beta_c}(\mathrm{d}A \mid \sigma)$ from the Edwards–Sokal law at β_c and define the auxiliary law

$$\hat{P}_N(\mathrm{d}\sigma, \mathrm{d}A) = P_{\beta_N}(\mathrm{d}\sigma)K_{\beta_c}(\mathrm{d}A \mid \sigma).$$

Its likelihood relative to the old joint law is precisely $\mathrm{d}P_{\beta_N}/\mathrm{d}P_{\beta_c}$, bounded above and below by positive constants. Thus the fixed contour events still give a positive decomposition of the new physical spin law. Their conditional exponential moments remain bounded, and their exceptional mass increases by at most a constant factor. The retained bond kernel is auxiliary; it need not be the Edwards–Sokal kernel at β_N . The same bounded spin likelihood directly preserves the mean-field positive phase moment bounds. Hence the isolated-copy $N^{3/2}$ thresholds persist under these shifts.

If independent momenta are included, their canonical partition factor is common to both spin phases and cancels from the weight ratio. Use the kinetic law at the same β_N as the spin reference. To check the effect on the complete kinetic law, its relative entropy against the law at β_c is

$$aN \left[\frac{\beta_c}{\beta_N} - 1 - \log \frac{\beta_c}{\beta_N} \right] = O(N^{-1}).$$

This follows either from the product Gaussian momenta or their Gamma energy law. Pinsker’s inequality gives total-variation convergence of the two kinetic laws. The mean energy changes by $O(1)$ and the $\sqrt{N}$ Gaussian fluctuation limit is unchanged. No globally bounded likelihood is asserted for the unbounded kinetic energy.

Use the resulting balanced reference for every copy and the same β_N in (5.3). Then Theorem 5.1 gives two canonical subsystem marginals, joint total-variation error $1/2$, and Proposition 5.2 gives one bit of mutual information. The reference being reproduced is the balanced canonical law at β_N , not the unbalanced law at β_c .

5.5 Phase events for the microscopic models

The hypotheses (5.1)–(5.2) ask for a partition of the state space into two phase events with conditionally tight $\sqrt{N}$ energy fluctuations. For the mean-field model of Section 4.1, take the ordered and disordered Voronoi cells; Lemma 4.2 gives conditional exponential moments on the $\sqrt{N}$ scale, hence tightness, with or without momenta. For the short-range model of Section 4.3, take the midpoint energy events $B_{o,N}$ and $B_{d,N}$ defined in Appendix A; their conditional standardized energies converge weakly by (A.13), hence are tight. The contour decomposition with its exceptional mass is not needed for the shared-bath theorems, because the centered weight (5.4) is bounded by one and cannot amplify rare configurations. After the balancing shift of Section 5.4, Lemma 5.4 preserves both partitions and their conditional limits.

6 Smooth reservoirs and the physical boundary law

6.1 What depends on the reservoir entropy

Let a reservoir induce a log likelihood $h_N(E)$ relative to a specified canonical law. For reservoir entropy $s_{B,N} = S_{B,N}/k_B$, $h_N(E) = \beta_N E + s_{B,N}(\mathcal{E}_N - E)$ up to a constant. Where temperature and positive heat capacity are defined by this entropy convention,

$$-h_N''(E) = -s_{B,N}''(\mathcal{E}_N - E) = \frac{\beta_B(\mathcal{E}_N - E)^2}{C_B(\mathcal{E}_N - E)/k_B}. \quad (6.1)$$

The proofs thus concern entropy curvature, with a heat-capacity interpretation when these thermodynamic derivatives exist.

Proposition 6.1 (Curvature necessity). *Suppose a positive decomposition (3.9) has two phases with positive limiting weights, centers $e_{-,N} < e_{+,N}$ separated by $\Delta_N \asymp N$, and conditional limits $(E - e_{i,N})/\sqrt{N} \Rightarrow \mathcal{N}(0, v_i)$ with $v_i > 0$. Let Q_N have likelihood proportional to e^{h_N} , with $h_N = -\infty$ outside an interval. On its interior assume h_N is twice continuously differentiable and $-h_N'' \geq k\kappa_N$ between the phase centers, for some $k > 0$ and a positive sequence $\kappa_N > 0$. Then $d_{TV}(P_N, Q_N) \rightarrow 0$ requires $\kappa_N N^{3/2} \rightarrow 0$.*

Proof. TV convergence gives $h_N(E) - \log Z_N \rightarrow 0$ in canonical probability on the feasible set. Choose good energies with standardized positions in $[1/2, 1]$ and $[3/2, 2]$ above the lower center, and $[-1, -1/2]$ below the upper center. Each interval has positive limiting probability. The selected energies $x_N < y_N < z_N$ are feasible, their log likelihoods differ from $\log Z_N$ by $o(1)$, and $y_N - x_N \asymp \sqrt{N}$, $z_N - y_N \asymp N$. Their convex hull is feasible and lies between the centers. Concavity of $h_N(E) + k\kappa_N E^2/2$ gives

$$h_N(y_N) - \frac{z_N - y_N}{z_N - x_N} h_N(x_N) - \frac{y_N - x_N}{z_N - x_N} h_N(z_N) \geq \frac{k\kappa_N}{2} (y_N - x_N)(z_N - y_N).$$

The left side tends to zero. This is the curvature version of Theorem 3.4's good-point chord argument. $\square$

Under the density hypotheses of Section 3.4, Proposition 3.7 supplies the converse when $h_N(e_{-,N}) = h_N(e_{+,N}) = 0$, h_N is globally concave on its feasible interval, and its curvature satisfies (3.19). A family with curvature comparable to κ_N therefore has the sharp $\kappa_N N^{3/2} \rightarrow 0$ criterion. This statement assumes that endpoint calibration exists; an arbitrary entropy need not realize it by varying only the composite energy. For exact separated Gaussian phases no interphase density hypothesis is needed: the chord and curvature upper bound give vanishing standardized endpoint slopes, and the tangent bound is uniformly integrable by Gaussian exponential moments. This gives the analogous criterion $\kappa_N \Delta_N s_N \rightarrow 0$ for a gap Δ_N and Gaussian fluctuation scale s_N .

6.2 Finite tilts at capacity proportional to $N^{3/2}$

Here $\beta_N \rightarrow \beta > 0$. Use the positive decomposition (3.9), with $w_{i,N} \rightarrow w_i > 0$, $w_- + w_+ = 1$, and $\Delta_N/N \rightarrow \ell > 0$. Write $\mathcal{E}_N^{\text{sec}}$ for (2.10). A Gaussian measure $G_i = \mathcal{N}(0, v_i)$ may have $v_i = 0$, interpreted as a point mass.

Theorem 6.2 (Physical boundary and finite calibration corrections). *Suppose $c_N/N^{3/2} \rightarrow \gamma \in (0, \infty)$ and*

$$X_{i,N} := (E - e_{i,N})/\sqrt{N} \Rightarrow G_i, \quad \sup_{N \geq N_0} \mathbb{E}_{P_{i,N}} e^{\eta |X_{i,N}|} < \infty \quad \text{for some } N_0 \text{ and } \eta > b := \frac{\beta^2 \ell}{2\gamma}. \quad (6.2)$$

Assume also

$$\delta_N \exp \left[\max_E h_N^{\text{sec}}(E) \right] \longrightarrow 0. \quad (6.3)$$

For $\mathcal{E}_N = \mathcal{E}_N^{\text{sec}} + d_N \sqrt{N}$, $d_N \rightarrow d$, put $b_- = b$, $b_+ = -b$, $D = \beta^2 \ell d / \gamma$, and

$$A_- = 0, \quad A_+ = D, \quad Z(d) = \sum_i w_i e^{A_i + b^2 v_i / 2}, \quad r_i(d) = \frac{w_i e^{A_i + b^2 v_i / 2}}{Z(d)}. \quad (6.4)$$

The reservoir phase probabilities, defined on the positive auxiliary decomposition, tend to $r_i(d)$; their standardized conditional fluctuations converge weakly to $G_i^b := \mathcal{N}(b_i v_i, v_i)$. Macroscopic energy separation makes these limiting probabilities observable from energy alone. On the full microscopic state space,

$$\lim_N d_{\text{TV}}(P_N, Q_N) = \frac{1}{2} \sum_i w_i \int_{\mathbb{R}} \left| \frac{e^{A_i + b_i z}}{Z(d)} - 1 \right| dG_i(z). \quad (6.5)$$

Proof. Normalize h_N at $e_{-,N}$. Uniformly on bounded standardized windows, the exact secant formula gives

$$h_N(e_{i,N} + \sqrt{N}z) = A_i + b_i z + o(1). \quad (6.6)$$

Indeed, with $q_N = \beta_N \Delta_N / c_N \rightarrow 0$, the secant endpoint derivatives are $\beta_N q_N / 2 + O(q_N^2)$ and $-\beta_N q_N / 2 + O(q_N^2)$. Their products with $\sqrt{N}$ tend to b and $-b$, and $N \sup |h_N''| = O(N/c_N) \rightarrow 0$ on these windows. Changing $\mathcal{E}_N$ by $d_N \sqrt{N}$ changes the upper-minus-lower endpoint value by

$$d_N \sqrt{N} c_N \left[(\mathcal{E}_N^{\text{sec}} - e_{+,N})^{-1} - (\mathcal{E}_N^{\text{sec}} - e_{-,N})^{-1} \right] + o(1) \longrightarrow D,$$

and changes each standardized endpoint slope by $o(1)$.

Concavity gives $h_N(E) \leq h_N(e_{i,N}) + h'_N(e_{i,N})(E - e_{i,N})$ on the feasible set. Hence $e^{h_N(E)} \leq C \exp[(b+\epsilon)|X_{i,N}|]$ for small fixed $\epsilon > 0$ and large N . Choose $p > 1$ with $p(b+\epsilon) < \eta$. Equation (6.2) gives uniform integrability, also for the absolute difference in (6.5). Cutoff points have zero weight and cause no difficulty. The maximum of h_N changes from the secant maximum by $O(1)$ for bounded d_N . Indeed, the maximum is at $E = \mathcal{E}_N - c_N / \beta_N$ and, when $h_N(e_{-,N}) = 0$, the derivative of its value with respect to $\mathcal{E}_N$ is $\beta_N - c_N / (\mathcal{E}_N - e_{-,N}) = O(N^{-1/2})$ in this range. Thus (6.3) removes the exceptional contribution. Weak convergence, (6.6), and uniform integrability now give $Z_N \rightarrow Z(d)$ and the Gaussian tilts. Finally use the exact identity $d_{\text{TV}}(P_N, Q_N) = \frac{1}{2} \mathbb{E}_{P_N} |e^{h_N} / Z_N - 1|$. This does not infer an integral limit from weak convergence alone. $\square$

The secant prescription $d = 0$ equalizes likelihoods at the centers, but its limiting integrated phase weights differ from w_i if the variances differ. If both variances equal v , the weights remain unchanged and

$$\lim_N d_{\text{TV}}(P_N, Q_N) = 2\Phi(b\sqrt{v}/2) - 1, \quad (6.7)$$

where Φ is the standard normal distribution function.

Corollary 6.3 (Compensating the phase probabilities). *Under these hypotheses, the calibration*

$$\mathcal{E}_N = \mathcal{E}_N^{\text{sec}} + \frac{\beta^2 \ell (v_- - v_+)}{8\gamma} \sqrt{N} + o(\sqrt{N}) \quad (6.8)$$

restores the original phase weights asymptotically, and

$$\lim_N d_{\text{TV}}(P_N, Q_N) = \sum_i w_i \{2\Phi(b\sqrt{v_i}/2) - 1\}. \quad (6.9)$$

Proof. The correction makes $D = b^2(v_- - v_+)/2$, so the integrated factors in (6.4) are equal. Within each phase the remaining change is the Gaussian translation by $b_i v_i$, with TV $2\Phi(b\sqrt{v_i}/2) - 1$. This is zero when $v_i = 0$. $\square$

Matching phase populations thus leaves a finite distortion of phase fluctuations. The correction does not assert exact finite- N matching.

6.3 Optimization over every composite energy

For finite signed measures let $\|\cdot\|_{\text{var}}$ be the total variation norm. Define the continuous convex function

$$F(r) = \frac{1}{2} \left\{ \|w_- G_- - r G_-^b\|_{\text{var}} + \|w_+ G_+ - (1-r) G_+^b\|_{\text{var}} \right\}, \quad 0 \leq r \leq 1. \quad (6.10)$$

Theorem 6.4 (Optimized physical boundary). *Under Theorem 6.2's hypotheses,*

$$\lim_N \inf_{\mathcal{E}: Z_N(\mathcal{E}) > 0} d_{\text{TV}}(P_N, Q_{N,\mathcal{E}}) = \min \left\{ \min_{0 \leq r \leq 1} F(r), w_-, w_+ \right\}. \quad (6.11)$$

For $w_- = w_+ = 1/2$ and $v_- = v_+ = v$, this is $\min\{2\Phi(b\sqrt{v}/2) - 1, 1/2\}$.

Proof. As d varies through $\mathbb{R}$, (6.4) realizes every strictly positive pair of phase weights. The boundary theorem therefore realizes $F(r)$ for every $r \in (0, 1)$; continuity supplies the upper bound by its infimum on $[0, 1]$. The additional upper bounds w_- and w_+ follow by centering the reservoir maximum at either phase, as in (2.7). Its weight lies in $[0, 1]$, tends to one in the selected phase since $N/c_N \rightarrow 0$, and tends to zero in the other since $\Delta_N^2/c_N \rightarrow \infty$. Positive decomposition and macroscopic separation give TV tending to the discarded phase weight.

For the lower bound consider a subsequence of arbitrary calibrations with TV limit strictly below $\min(w_-, w_+)$. Let R_N be their normalized likelihood. The overlap in phase i is $O_{i,N} = w_{i,N} \mathbb{E}_{P_{i,N}} \min(R_N, 1)$. Since $d_{\text{TV}}(P_N, Q_N) = \mathbb{E}_{P_N}(1 - R_N)_+$, each phase loss is at most TV; hence $O_{i,N} \geq w_{i,N} - d_{\text{TV}}(P_N, Q_N)$. Both overlaps stay positive. Tightness, the overlap bound from $R_N < \epsilon$, and $P_N(R_N > M) \leq 1/M$ allow selection of feasible points $x_N = e_{-,N} + O(\sqrt{N})$ and $y_N = e_{+,N} + O(\sqrt{N})$ with $R_N(x_N), R_N(y_N) \in [\epsilon, M]$, for fixed positive ϵ, M . Set $a_N = \log R_N(y_N) - \log R_N(x_N) = O(1)$ and $D_N = y_N - x_N \asymp N$. The exact likelihood solves to

$$\mathcal{E}_N = x_N + \frac{D_N}{1 - \exp[-(\beta_N D_N - a_N)/c_N]}. \quad (6.12)$$

A bounded a_N changes this by $O(c_N/D_N) = O(\sqrt{N})$ from the secant calibration for x_N, y_N . Uniformly on these windows that calibration is

$$\frac{c_N}{\beta_N} + \frac{x_N + y_N}{2} + \frac{\beta_N(y_N - x_N)^2}{12c_N} + O\left(\frac{(y_N - x_N)^4}{c_N^3}\right).$$

Moving its endpoints by $O(\sqrt{N})$ changes it by $O(\sqrt{N})$. Thus $(\mathcal{E}_N - \mathcal{E}_N^{\text{sec}})/\sqrt{N}$ is bounded. Pass to a convergent subsequence and apply Theorem 6.2; the limit is $F(r) \geq \min F$. Sequences with limiting error at least $\min(w_-, w_+)$ already obey the required bound. Apply this argument to calibrations within $1/N$ of the infimum; no existence of minimizers is assumed. The upper and lower bounds coincide, so the limit in (6.11) exists and has the stated value. In the balanced equal-variance case reflection sends r to $1 - r$, so convexity minimizes F at $1/2$, giving the normal formula. $\square$

The optimization is a scalar calculation. Set $r_- = r$, $r_+ = 1 - r$, and $d_i = b\sqrt{v_i}$. For $d_i > 0$ the contribution of phase i to the positive part of reservoir minus reference, $r_i G_i^b - w_i G_i$, is

$$r_i \Phi\left(\frac{\log(r_i/w_i)}{d_i} + \frac{d_i}{2}\right) - w_i \Phi\left(\frac{\log(r_i/w_i)}{d_i} - \frac{d_i}{2}\right). \quad (6.13)$$

Their sum is $F(r)$ because total positive and negative masses agree. Use zero for $r_i = 0$, by continuity, and $(r_i - w_i)_+$ when $d_i = 0$. Equation (6.13) follows by integrating on the half-line where the tilted Gaussian density exceeds the weighted original density. If both $d_i > 0$, differentiation gives

$$F'(r) = \Phi\left(\frac{\log(r/w_-)}{d_-} + \frac{d_-}{2}\right) - \Phi\left(\frac{\log((1-r)/w_+)}{d_+} + \frac{d_+}{2}\right).$$

This is strictly increasing from a negative to a positive value. Its unique zero, equivalently equality of the two CDF arguments, determines the unique minimizer. For equal positive variances it is $r = w_-$, even with unequal target weights. Thus the optimized limit then simplifies to $\min\{2\Phi(b\sqrt{v}/2) - 1, w_-, w_+\}$.

6.4 An exact physical gain and its interfacial scale

At a fraction θ of the gap, the endpoint-normalized physical secant likelihood has the exact form

$$h_N(e_{-,N} + \theta\Delta_N) = c_N [\theta q_N + \log(1 - \theta + \theta e^{-q_N})], \quad q_N = \frac{\beta_N \Delta_N}{c_N}. \quad (6.14)$$

For $q_N \rightarrow 0$, uniformly on $0 \leq \theta \leq 1$,

$$\begin{aligned} h_N(e_{-,N} + \theta\Delta_N) &= \frac{\beta_N^2 \Delta_N^2}{2c_N} \theta(1 - \theta) \\ &\quad + \frac{\beta_N^3 \Delta_N^3}{6c_N^2} \theta(1 - \theta)(2\theta - 1) + O(\Delta_N^4/c_N^3). \end{aligned} \quad (6.15)$$

This follows by expanding the logarithm of the Bernoulli transform in (6.14); bounded θ gives a uniform remainder. The maximum lies inside the two centers because the endpoint slopes have opposite signs. Equation (6.15) therefore proves (6.16). At $c_N \asymp N^{3/2}$ the cubic term can be of order one, so the quadratic approximation does not determine all subleading weights.

An interface with surface cost of order $L^{d-1} = N^{(d-1)/d}$ competes with a maximal bath gain of order N^2/c_N . Their balance occurs at $c_N \asymp N^{1+1/d}$. For $d > 2$ this is below the $N^{3/2}$ fluctuation threshold. For $d = 2$ they coincide. This comparison explains both the exponent $\alpha = 1/2$ in the density criterion and the restricted two-dimensional boundary range (6.17). It is a scale comparison; a surface bound alone does not identify a favored droplet or slab.

6.5 Microscopic applications of the boundary law

The boundary law holds for the mean-field model of Theorem 4.1, for every $\gamma > 0$ and kinetic parameter $a \geq 0$. The occupation bound in Lemma 4.2 is Gaussian on the $\sqrt{N}$ scale, so it bounds $\mathbb{E}e^{t|X_{i,N}|}$ for every fixed t and sufficiently large N . The independent Gamma contribution has the same property by its exact transform. There is no exceptional component. The variances are (4.10); for pure spins $v_+ = 0$, which is allowed here. A continuous kinetic energy is unnecessary for the boundary theorem or its optimization.

For the short-range model of Theorem 4.3, the boundary law and its optimization hold for every $\gamma > 0$ in fixed spatial dimension $d \geq 3$, and for all sufficiently large fixed γ in $d = 2$. The additional

argument is needed because a small exponential moment does not justify every finite tilt. The cutoff-free restricted-partition window in Lemma A.1 has width η_0/L . When $d > 2$, every fixed multiple of $N^{-1/2} = L^{-d/2}$ eventually lies in that window. The second derivative bound (A.15) and center estimate (A.16) therefore bound exponential moments of the standardized bond count at every fixed parameter. The inverse-temperature to bond-fugacity change has bounded derivatives and preserves this conclusion. The conditional binomial estimate and Cauchy–Schwarz argument in Lemma A.2 transfer these bounds to the actual spin energy. The contour-phase weak limits agree with (A.13): the already proved small exponential moment bounds the conditional probability of the opposite macroscopic midpoint phase by $Ce^{-c\sqrt{N}}$, and the exceptional contour probability vanishes. Positive limiting phase weights then identify the conditional laws.

The exceptional probability is at most $Ce^{-\tau L^{d-1}}$. The exact secant likelihood satisfies

$$\frac{\max_E h_N^{\text{sec}}(E)}{\sqrt{N}} \rightarrow \frac{\beta^2 \ell^2}{8\gamma}. \quad (6.16)$$

For $d > 2$ the surface exponent dominates $\sqrt{N}$, giving (6.3) for every γ . For $d = 2$, let $\eta_* > 0$ be a valid spin-phase moment coefficient and $\tau > 0$ a valid exceptional surface constant. It suffices that

$$\frac{\beta^2 \ell}{2\gamma} < \eta_*, \quad \frac{\beta^2 \ell^2}{8\gamma} < \tau. \quad (6.17)$$

These hold for all sufficiently large γ . No assertion about all $\gamma > 0$ or small-capacity microscopic morphologies in two dimensions follows from these estimates.

An alternative sufficient input is the continuous density envelope (3.18) with $\alpha > 1/2$, together with (3.17). These also give the boundary law and finite corrections for every fixed γ . On the interior put $x = \min_i |E - e_{i,N}|$. Concavity gives $h_N^{\text{sec}}(E) \leq Cx/\sqrt{N}$ at this boundary. With a finite calibration correction, the endpoint values are bounded but need not both vanish, so the bound instead reads $h_N(E) \leq C_0 + C_1 x/\sqrt{N}$ with constants independent of N . For $x \geq R\sqrt{N}$ the ratio of the linear term to x^2/N is at most C_1/R , and its ratio to x^α is at most $C_1 N^{1/2-\alpha}$. Choose R large and then N large to absorb this term in the envelope. The offset contributes only the bounded prefactor e^{C_0} to the tail integral and does not affect its vanishing limit as $R \rightarrow \infty$. Exterior to the centers the secant weight is at most one. For finite corrections, concavity and the endpoint slope signs instead bound exterior weights by a constant, which suffices. Weighted tail control and local density convergence give the same integral formula. At $\alpha = 1/2$ the second ratio need not vanish. The capillarity model in Appendix C explains this distinction.

7 Deterministic numerical illustrations

Figure 1 evaluates the mean-field microscopic model with $J = 1$, $\beta = 4 \log 2$, and $a = 1/2$. Occupation triples are enumerated modulo color permutations, and the Gamma and Beta conditional energy integrals are evaluated analytically through their distribution functions, as derived in Section 4. Thus the plotted TV is the error of the full spin-momentum law, not merely the spin-configuration marginal. No Monte Carlo sampling or Gaussian approximation enters the finite- N points.

At $c_N = 0.5N^{3/2}$ the predicted limit is approximately 0.157392; the exact $N = 12000$ value is 0.148341. At $c_N = 2N^{3/2}$ the corresponding values are 0.037986 and 0.037304. These are secant-calibrated errors, not results of a finite- N optimization. The $N^{7/4}$ sequence illustrates decay above the threshold; the $N^{5/4}$ sequence illustrates persistent error below it. The finite range of sizes does not establish an asymptotic exponent by itself. The theorem provides that conclusion.

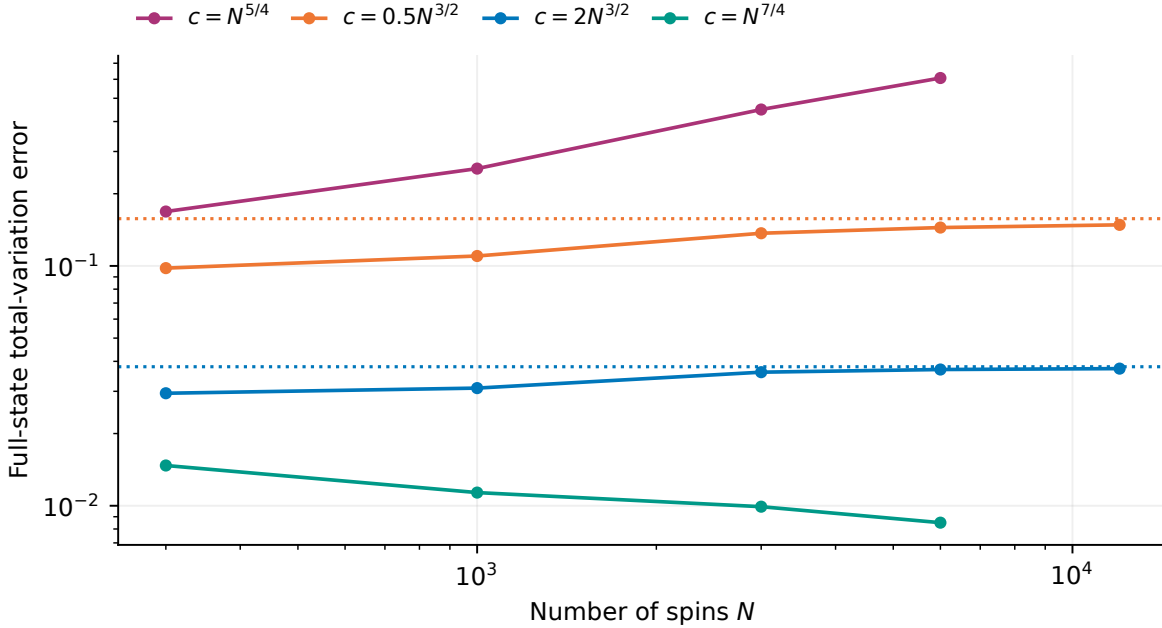


Figure 1: Exact finite-system mean-field Potts calculations with the physical secant bath. Points are occupation sums and Gamma/Beta CDF evaluations, connected to guide the eye. Dotted lines are the proved secant boundary limits for $c_N = \gamma N^{3/2}$, with $\gamma = 0.5$ and 2. The largest size, $N = 12000$, was computed for these two boundary sequences; the other sequences extend to $N = 6000$. These are mean-field data and do not test a short-range interfacial law.

These sizes are also far from the asymptotic regime of the phase decomposition itself. The saddle of the rate function (4.3) at $(1/2, 1/4, 1/4)$ lies on the Voronoi boundary between the ordered and disordered cells, and its rate excess is $I(1/2, 1/4, 1/4) - I_{\min} = \log 3 - \frac{19}{12} \log 2 \simeq 1.1 \times 10^{-3}$. The interphase valley is therefore suppressed only by $\exp(-0.0011N)$, which is of order one below $N \simeq 10^3$. Exact occupation sums for the pure-spin model show the slow approach. Here Voronoi distance is Euclidean in the three occupation fractions; ties between ordered and disordered cells are assigned to the ordered phase. The combined ordered event is therefore $\max_j n_j \geq N/2$. With this convention, the Voronoi-conditional ordered variance divided by N is 1.02 at $N = 3000$ and 0.78 at $N = 12000$, against the limit $v_- \simeq 0.733$ from (4.10), while the ordered weight is 0.528 and 0.529 against $w_- \simeq 0.530$ from (4.9). The limiting formulas plotted as dotted lines use the limiting constants; the finite-size points are exact.

Figure 2 displays a different object: the proved limiting formula for balanced phases of equal variance, rather than microscopic finite-size enumeration. Writing $\lambda = b\sqrt{v}$, preserving the phase weights gives $2\Phi(\lambda/2) - 1$, whereas optimizing the total energy gives $\min\{2\Phi(\lambda/2) - 1, 1/2\}$. The second branch is attained by abandoning one phase. This example explains why an error report should include both the TV value and the resulting phase populations.

The accompanying source bundle retains the finite-system data, figure code, and independent checks. Small systems are checked against explicit enumeration of all 3^N labeled spin states for $1 \leq N \leq 8$. At $N = 12$, direct absolute-density integration verifies the Gamma/Beta crossing formula for two kinetic shapes. Separate normal-density quadratures verify the weighted CDF expression (6.13) and its optimizer. Floating-point calculations are numerical checks, not interval certificates or substitutes for the proofs. The reproducibility instructions distinguish quick checks

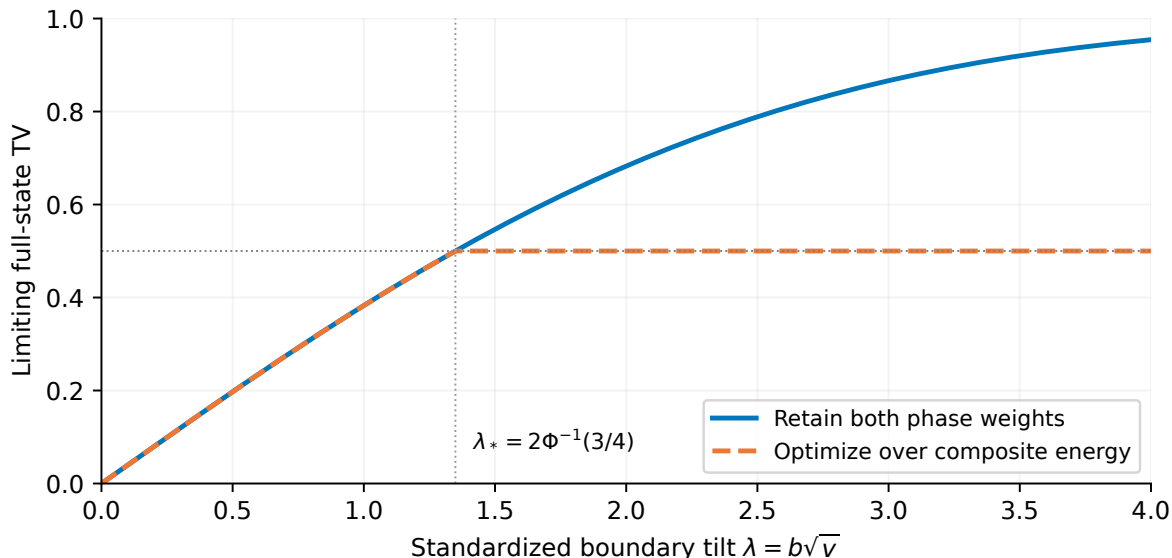


Figure 2: The physical boundary law for balanced equal-variance phases under Theorem 6.2’s moment and exceptional-mass hypotheses. This is an evaluation of the limiting formula, not finite-size Potts data. At $\lambda_* = 2\Phi^{-1}(3/4) \simeq 1.34898$, an optimizer that retains both phase populations and one that discards a phase have the same limiting error. The same curve holds for the exact scalar Gaussian model of Proposition B.1.

and figure regeneration from the more expensive full occupation enumeration through $N = 12000$.

8 Discussion and conclusions

A reservoir requirement becomes precise only after specifying what must be reproduced. The complete canonical probability law imposes constraints on both phase populations and phase fluctuations. For the physical power-law reservoir, a weak energy limit with at least three support points already forces capacity growing faster than the square of its scale, even after arbitrary total-energy tuning. Two separated energy phases permit an endpoint calibration that reduces the obstruction to the product of latent-energy gap and fluctuation width. Positive phase tails are then needed for sufficiency, because a small canonical interfacial probability can be amplified by the bath.

These statements have microscopic content. The mean-field and short-range Potts results verify the relevant hypotheses for interacting Hamiltonians, including plain spin models. The short-range proof uses positive contour measures and an explicit bond-to-energy transfer; it does not replace a signed partition remainder by a probability error. At the $N^{3/2}$ boundary, the phase measures acquire finite exponential tilts. Their probabilities can be compensated by an additional physical calibration, but this does not remove the fluctuation distortion. The optimization theorem also shows that the smallest TV error can come from discarding a phase. Phase populations remain an essential diagnostic even when the metric itself has been optimized.

Shared reservoirs require a further distinction between separate and joint tests. Two independent copies can each have a canonical full marginal while their phase assignments remain anticorrelated and their microscopic mutual information approaches one bit. This is an equilibrium effect of the stated common reservoir. It does not require direct subsystem interaction, and it does not extend without modification to arbitrary interacting spatial subregions. The full information limit follows

from a separate relative-entropy argument; it cannot be inferred from TV continuity on growing microscopic spaces.

For finite-system thermodynamics, these results separate capacity, calibration, and the measured observable. The heat-capacity convention matters at finite size, as finite ideal-gas contact calculations also emphasize [11]. Smooth positive-capacity entropies share the curvature mechanism when an appropriate calibration and tail control are available. Barrier accuracy is stronger in a different sense: at specified midpoint and endpoint energies the exact logarithmic barrier lowering can remain large even while full TV tends to zero. Conversely, macroscopic energy concentration or a small histogram Euclidean norm does not establish full-law accuracy.

The proved scope is an additive equilibrium coupling to the stated bath, fixed positive limiting phase weights, and the explicit fluctuation and tail hypotheses of each theorem. The short-range result requires a fixed number of colors $q \geq q_0(d)$. Its boundary law holds for every fixed positive capacity coefficient in dimensions above two and for sufficiently large coefficients in two dimensions. The square-torus interfacial calculation is a stated capillarity model, not a derivation of microscopic morphology. The appendices keep this separation explicit: the contour proof verifies a microscopic hypothesis, whereas the Gaussian, capillarity, and linear-capacity appendices analyze stated models exactly.

A Positive contour measures and the short-range theorem

This appendix proves Theorem 4.3. Its source estimates, phase definitions, and energy observable are kept explicit because a signed partition-function approximation alone does not control a positive reweighted probability measure.

The color threshold $q_0(d)$ is fixed once, here. It is the largest of: the thresholds required by the cited finite-size scaling and contour results [4, 3] on a fixed temperature interval around coexistence; the value $2d + 1$, used for the variance lower bound in (A.14); and the value needed for the cluster-expansion criterion of [3] to hold with the extra factor $e^{\rho m \gamma}$ in (A.19) for a fixed $\rho > 0$. Every “sufficiently large q ” below refers to this single threshold.

A.1 Published finite-volume inputs and normalization

We use the following established contour results, with source locations given so that the deductions can be distinguished from their inputs. Theorem 1 of Borgs, Kotecký, and Miracle-Solé [4] gives six-times differentiable dimensionless free energies ψ_o, ψ_d and a fixed real interval containing β_c on which

$$Z_N^{\text{spin}}(\beta) = qe^{-N\psi_o(\beta)} + e^{-N\psi_d(\beta)} + O(e^{-bL})e^{-N \min_i \psi_i(\beta)}. \quad (\text{A.1})$$

The constants are uniform on a sufficiently small fixed compact interval. The source states its error as $O(q^{-bL})$; for fixed q this is $O(e^{-bL})$ with a different constant b , which is the form used here. At coexistence the free energies are equal and

$$u_i = \psi'_i(\beta_c), \quad u_o < u_d, \quad \sigma_i^2 = -\psi''_i(\beta_c). \quad (\text{A.2})$$

The ordered branch is stable above β_c , and the disordered branch below. The source Hamiltonian in its equation (6), with $J = 1$, is exactly (4.19). The notation here uses $\psi_i = \beta f_i$ relative to that source. No complex-temperature continuation of (A.1) is assumed.

For the positive decomposition, put $p = 1 - e^{-\beta}$ and use bond sets $A \subseteq \mathcal{B}_L$. Their random-cluster weights are

$$w_\beta(A) = p^{|A|}(1-p)^{dN-|A|}q^{k(A)}, \quad Z_N^{\text{RC}}(\beta) = \sum_A w_\beta(A) = e^{-\beta dN} Z_N^{\text{spin}}(\beta). \quad (\text{A.3})$$

This normalization follows by expanding $\prod_{\{x,y\}}[1 + (e^\beta - 1)\mathbf{1}_{\{\sigma_x = \sigma_y\}}]$. It fixes the additive energy convention independently of conventions used in contour references.

Throughout the short-range proof, contours, interiors and exteriors, compatibility, phase events, restricted partition functions, and activities are those of Borgs, Chayes, and Tetali [3] (BCT), Definitions 4.1–4.4 and 5.4, equations (6.1)–(6.5) and (6.13)–(6.22). In particular, BCT defines the exterior of a contour as the component containing a compatible interface, if such an interface exists; otherwise it chooses the larger component, with a fixed tie rule. This convention matters for large contours on the torus.

Geometrically, the construction thickens the occupied bonds and separates ordered and disordered regions by their boundary components. Components with zero winding vector are contours; the remaining components are interfaces. In the absence of interfaces, the common exterior of the contours has one label, ordered or disordered, which defines the positive event $\mathcal{O}_L$ or $\mathcal{D}_L$. Configurations with at least one interface form the tunneling event $\mathcal{T}_L$. Thus these events partition the bond configurations and are independent of β . Let $Z_{o,L}$ denote the ordered contribution with its color multiplicity removed, and $Z_{d,L}$ the disordered contribution. BCT (6.1)–(6.5) give exactly

$$Z_N^{\text{RC}} = qZ_{o,L} + Z_{d,L} + Z_{t,L}. \quad (\text{A.4})$$

At β_c , BCT Lemma 6.1 bounds the tunneling probability by $Ce^{-bL^{d-1}}$. The later construction in [2], Definition 5, uses a modified exterior convention; we do not identify its finite-volume phase events with those used here or import estimates between them.

We will use the following specific parts of the original contour construction. Write $m_\gamma = \|\gamma\|$ for the contour size. Its diameter obeys

$$\text{diam } \gamma \leq L, \quad \text{diam } \gamma \leq m_\gamma/2, \quad |\text{Int } \gamma \cap \Lambda_L| \leq \frac{1}{2}m_\gamma \text{diam } \gamma. \quad (\text{A.5})$$

These are the definition of diameter and Lemma 5.7 of [3]; they hold for all $d \geq 2$. The positive restricted partition functions have the polymer form

$$Z_{i,L} = e^{-e_i(\beta)N} \sum_{\Gamma \text{ compatible}} \prod_{\gamma \in \Gamma} K_i(\gamma; \beta). \quad (\text{A.6})$$

Here the explicit ground coefficients are $e_o = -d \log(1 - e^{-\beta})$, $e_d = d\beta - \log q$, and $\kappa = \frac{1}{2} \log(e^\beta - 1)$. The activities are $e^{-\kappa(\beta)m_\gamma}$ times a ratio of positive interior partition functions, with a factor q in the disordered case; see equations (6.18)–(6.22) of that source. The coefficients e_i and κ have bounded derivatives on the fixed temperature interval considered here.

Appendix A defines truncated activities

$$K'_i(\gamma; \beta) = \min\{K_i(\gamma; \beta), e^{-\tau(\beta)m_\gamma}\}, \quad (\text{A.7})$$

where BCT (A.3) uses $\tau(\beta) = \beta/8 - \beta/20 + 1$. This positive exponential cap ensures uniform convergence of the cluster expansion when q is sufficiently large. Its infinite-volume free energies will be denoted f_i^{tr} , to distinguish them from the ψ_i in (A.1). If $a_i = f_i^{\text{tr}} - \min_j f_j^{\text{tr}}$, Lemma A.1(i) states

$$a_i(\beta) \text{diam } \gamma \leq c_0\beta \implies K_i(\gamma; \beta) = K'_i(\gamma; \beta). \quad (\text{A.8})$$

Moreover, equation (A.8) on the full torus gives

$$|\log Z'_{i,L}(\beta) + Nf_i^{\text{tr}}(\beta)| \leq CNe^{-bL}. \quad (\text{A.9})$$

The assumptions of Appendix A cover a fixed neighborhood on both sides of β_c , even though the main-text Lemma 6.3 is stated on the ordered side. To identify the stable sides within this same construction, BCT Lemma A.3 says that the spontaneous magnetization is positive exactly when $a_o = 0$. BCT define their transition point β_0 as the onset of positive magnetization, whereas β_c above is the crossing $\psi_o = \psi_d$ of the BKMS free energies. The two coincide: for $q \geq q_0(d)$ the model has a unique, disordered Gibbs state below the large- q transition point and q ordered extremal states above it [23, 24], and the BKMS expansion (A.1) is derived from the same Pirogov–Sinai construction, in which the phase with the smaller ψ_i is the stable one at each β . Hence the magnetization vanishes below β_c and is positive above it, so $a_d = 0$ below and $a_o = 0$ above. Continuity gives $f_o^{\text{tr}} = f_d^{\text{tr}}$ at coexistence.

The minimum $f^{\text{tr}} = \min_i f_i^{\text{tr}}$ is the physical thermodynamic random-cluster free energy. This identification can be made directly in the original convention: BCT Lemma A.1(ii), on the full torus, bounds each $Z_{i,L}$ above by $\exp[-Nf^{\text{tr}} + O(Ne^{-bL})]$, while (A.9) supplies the corresponding lower bound for any minimizing phase. The interface-network sum (6.26)–(6.27) bounds $Z_{t,L}$ by $\exp[-Nf^{\text{tr}} - bL^{d-1}]$. That sum uses only the bounds from Appendix A and the large- q contour counting inequalities, which hold throughout the fixed neighborhood under consideration; shrink the neighborhood if necessary so that $q \leq e^{2d\kappa+d}$ there, as required in (6.27). The sum does not

require the ordered phase to minimize. Combining these estimates with (A.4) identifies the limiting pressure. Equation (A.3) and the stable branch of (A.1) consequently give

$$f_i^{\text{tr}}(\beta) = \psi_i(\beta) + d\beta \quad \text{on the stable side of phase } i. \quad (\text{A.10})$$

We do not identify the two metastable constructions away from their stable sides or assume differentiability of the minimum cutoff.

A.2 Weak phase energy limits and nonzero variances

Applying (A.1) at $\beta_c + t/N$ and using the boundedness of U_N/N gives

$$U_N/N \Rightarrow \frac{q}{q+1}\delta_{u_o} + \frac{1}{q+1}\delta_{u_d}. \quad (\text{A.11})$$

Indeed the ratios of partition functions are Laplace transforms and converge for every fixed real t to those of the displayed bounded law. Define the actual spin events $B_{o,N} = \{U_N/N < (u_o + u_d)/2\}$ and $B_{d,N} = B_{o,N}^c$. Their limiting probabilities are the two weights in (A.11).

For $s > 0$, use (A.1) at $\beta_c + s/\sqrt{N}$. The ordered branch dominates, and its quadratic Taylor expansion gives

$$\mathbb{E}_{\beta_c} \exp[-s(U_N - Nu_o)/\sqrt{N}] \longrightarrow \frac{q}{q+1} e^{\sigma_o^2 s^2/2}. \quad (\text{A.12})$$

The part on $B_{d,N}$ is at most $e^{-s(u_d - u_o)\sqrt{N}/2}$ and vanishes. Thus this is also the transform limit of the positive subprobability measure restricted to $B_{o,N}$. The disordered phase uses the opposite sign and $\beta_c - s/\sqrt{N}$.

For clarity, a one-sided transform limit suffices here. Tilt the ordered subprobability law of $x = (U_N - Nu_o)/\sqrt{N}$ by $e^{-s_0 x}$ for any fixed $s_0 > 0$ and normalize it. Ratios of (A.12) at $s_0 - t$ and s_0 , for $|t| < s_0$, give convergence of its moment-generating function to that of $\mathcal{N}(-\sigma_o^2 s_0, \sigma_o^2)$. The ordinary moment-generating function convergence theorem gives weak convergence of these tilted probabilities. Untilting against compactly supported continuous functions gives vague convergence of the original positive measures. Their known total masses converge to the recovered Gaussian mass, so vague convergence is weak convergence. Hence

$$(U_N - Nu_i)/\sqrt{N} \mid B_{i,N} \Rightarrow \mathcal{N}(0, \sigma_i^2), \quad i = o, d. \quad (\text{A.13})$$

The variance is initially allowed to be zero; nonnegativity follows from convexity of the limiting log transforms.

Both variances are in fact strictly positive. Here the requirement $q > 2d$ in the choice of $q_0(d)$ is used. Choose an independent vertex set S of size at least $N/(2d+1)$ and condition on all spins outside it. The remaining spins are independent. For $x \in S$, let $n_{x,a}$ count the neighbors of color a . Then $\sum_a n_{x,a} = 2d$, and the conditional color probabilities on a compact inverse-temperature interval with upper endpoint $\beta_{\max}$ obey $p_{x,a} \geq q^{-1}e^{-2d\beta_{\max}}$. If $q > 2d$, two neighbor counts differ by at least one. Since $\text{Var}_p(n) = \sum_{a < b} p_a p_b (n_a - n_b)^2$, conditional variance gives

$$\text{Var}_\beta U_N \geq v_* N, \quad v_* := \frac{e^{-4d\beta_{\max}}}{(2d+1)q^2} > 0. \quad (\text{A.14})$$

Thus $N^{-1} \log Z_N^{\text{spin}} - v_* \beta^2/2$ is convex. Its pointwise thermodynamic limit is convex as well. On each open stable side that limit equals $-\psi_i(\beta) - v_* \beta^2/2$, so $-\psi_i'' \geq v_*$. Continuity of ψ_i'' extends the bound to β_c , proving $\sigma_i^2 \geq v_* > 0$. This argument passes through the stable thermodynamic branches; it does not infer a phase variance from the much larger total coexistence variance.

A.3 Uniform moments from positive contour measures

We next establish the inward fluctuation control needed for sufficiency. The finite-volume argument is useful because the minimum in (A.7) need not be twice differentiable. The proof requires only a Lipschitz estimate for its limiting free energy and then removes every cutoff in a small real temperature interval.

Lemma A.1 (Restricted contour partition derivatives). *For the fixed sufficiently large q above, there are constants $\eta, C > 0$ such that, for $i = o, d$ and all sufficiently large L ,*

$$\left| \frac{d^2}{d\beta^2} \log Z_{i,L}(\beta) \right| \leq CN, \quad |\beta - \beta_c| \leq \eta/L. \quad (\text{A.15})$$

On this interval $Z_{i,L} = Z'_{i,L}$, and

$$\frac{d}{d\beta} \log[e^{d\beta N} Z_{i,L}(\beta)] \Big|_{\beta=\beta_c} = -Nu_i + O(\sqrt{N}). \quad (\text{A.16})$$

Proof. We supply the derivative bounds underlying the use of the cluster expansion. For a general contour interior $\Lambda = \text{Int } \gamma$, use the positive matching-label sums in BCT (6.13)–(6.14). A summand indexed by an admissible family Γ has logarithm

$$F_\Gamma(\beta) = -e_o(\beta)n_o(\Gamma) - e_d(\beta)n_d(\Gamma) - \kappa(\beta)M(\Gamma) + c_\Gamma \log q, \quad (\text{A.17})$$

where $n_o + n_d = |\Lambda \cap \Lambda_L|$, the nonnegative integers n_o, n_d count vertices with each label, and $M(\Gamma) = \sum_{\gamma' \in \Gamma} m_{\gamma'}$ counts internal contour intersections. The integer c_Γ counts ordered components, with the constant subtraction of one when the ordered color multiplicity is removed. All these quantities and the set of admissible families are independent of β .

To bound $M(\Gamma)$, recall BCT's construction from cubes of side $1/2$ and its definition of contour size as the number of intersections with lattice edges, Section 4 and equation (4.5). Each contour surface is a union of faces of the half-integer cube lattice, so it meets a given lattice edge in at most a fixed number C'_d of points, and pairwise compatible contours are disjoint, so the intersections counted by $M(\Gamma)$ are distinct points on distinct edges or on distinct contours. A contour of Γ lies in the closure of Λ , so the edges it meets either have an endpoint in $\Lambda \cap \Lambda_L$, of which there are at most $2d|\Lambda \cap \Lambda_L|$, or cross the bounding contour γ , of which there are at most m_γ by the definition of contour size. Consequently

$$M(\Gamma) \leq C'_d(2d|\Lambda \cap \Lambda_L| + m_\gamma) \leq C_d(|\Lambda \cap \Lambda_L| + m_\gamma) =: C_d V_\gamma.$$

The bounded derivatives of e_o, e_d, κ now give $|F'_\Gamma| + |F''_\Gamma| \leq C V_\gamma$. This argument applies to the matching-label sums even when the interior wraps around the torus; it requires no interpretation as an unrestricted random-cluster partition function on a simply connected region.

For a positive sum $S = \sum_\Gamma e^{F_\Gamma}$, direct differentiation gives

$$(\log S)' = \mathbb{E}_S F'_\Gamma, \quad (\log S)'' = \mathbb{E}_S F''_\Gamma + \text{Var}_S F'_\Gamma.$$

Hence $|(\log S)'| \leq C V_\gamma$ and $|(\log S)''| \leq C(V_\gamma + V_\gamma^2)$. Apply these bounds to the numerator and denominator in BCT's activity ratios (6.18) and (6.22). Since $V_\gamma = O(m_\gamma^2)$ by (A.5), this gives the uniform real-parameter bounds

$$|\partial_\beta \log K_i(\gamma)| \leq C m_\gamma^2, \quad |\partial_\beta^2 K_i(\gamma)| \leq C m_\gamma^4 K_i(\gamma). \quad (\text{A.18})$$

These inequalities do not require the interior phase to be stable.

Here are the precise summability estimates used to differentiate. For a cluster $\mathcal{C}$, write $m(\mathcal{C}) = \sum_{\gamma \in \mathcal{C}} m_\gamma$ and let $\Phi(\mathcal{C})$ denote its Ursell coefficient, including the factor for repeated polymers. The contour counting and incompatibility estimates (A.5)–(A.6) of [3] give, for some fixed $\rho > 0$,

$$\sum_{\mathcal{C}} |\Phi(\mathcal{C})| \prod_{\gamma \in \mathcal{C}} K'_i(\gamma; \beta) e^{\rho m(\mathcal{C})} \leq CN. \quad (\text{A.19})$$

The factor $e^{\rho m_\gamma}$ is absorbed by the choice of $q_0(d)$: the criterion (A.5)–(A.6) holds with a slack $e^{(c\beta+1)m_\gamma}$ that grows with q , so a fixed $\rho < c\beta + 1$ leaves the criterion satisfied on the temperature interval. To sum over clusters, use the device of BCT's proof of Lemma A.2 (their Appendix A.3): every contour in Λ_L is incompatible with at least one of the N contours of size two obtained from configurations with a single disordered edge in a fixed direction. Every cluster therefore contains a contour incompatible with one of these N fixed contours, and summing the incompatibility bound (A.6) over them, with the polynomial factor absorbed in the exponential slack, proves (A.19). Overcounting clusters that are incompatible with several fixed contours only increases the upper bound.

On any compact real temperature interval, the minimum in (A.7) is absolutely continuous. Almost everywhere, its logarithmic derivative is that of one of its two positive branches; both obey $|\partial_\beta K'_i| \leq Cm_\gamma^2 K'_i$. Termwise first differentiation is therefore justified by (A.19), since a differentiated cluster is bounded by $Cm(\mathcal{C})^2$ times its absolute weight. Equivalently, integrate these almost-everywhere derivatives over the parameter interval and use the same majorant to interchange sum and integral. Including the ground term in (A.6) gives $|\partial_\beta \log Z'_{i,L}| \leq CN$ almost everywhere. Hence the finite-volume free energies are uniformly Lipschitz. Their pointwise limits f_i^{tr} are Lipschitz with the same constant.

Since $a_i(\beta_c) = 0$ and a_i is the difference of two functions with Lipschitz constant C , it follows that $a_i(\beta) \leq 2C|\beta - \beta_c|$. Choose $\eta > 0$ small enough that $2C\eta < c_0 \inf \beta$ on the fixed interval. Every torus contour has diameter at most L , so (A.8) now removes *all* cutoffs on $|\beta - \beta_c| \leq \eta/L$. The finite-volume activities on this interval are their original smooth positive functions. A second derivative of a cluster product is bounded by $Cm(\mathcal{C})^4$ times its absolute weight, by (A.18). The exponential factor in (A.19) absorbs this polynomial uniformly. Twice differentiating the uniformly convergent cluster expansion, and adding the ground term, proves (A.15). This step differentiates no minimum cutoff.

For the center, set $F_{i,L}(\beta) = \log[e^{d\beta N} Z_{i,L}(\beta)]$ and take $t_N = \eta/(2\sqrt{N})$ with the sign pointing to the stable side of phase i . Since $d \geq 2$, $|t_N| \leq \eta/L$. Equations (A.9) and (A.10) give, both at β_c and at $\beta_c + t_N$, $F_{i,L}(\beta) = -N\psi_i(\beta) + O(Ne^{-bL})$. The second-derivative bound and a stable-side finite difference imply

$$F'_{i,L}(\beta_c) = -N \frac{\psi_i(\beta_c + t_N) - \psi_i(\beta_c)}{t_N} + O(N|t_N|) + O(Ne^{-bL}/|t_N|) = -Nu_i + O(\sqrt{N}).$$

The exponentially small error is negligible for fixed d , proving (A.16). $\square$

Lemma A.2 (Transfer from bond fluctuations to spin energy). *In the Edwards–Sokal joint law at β_c , the conditional spin energy on each of the positive bond phase events obeys*

$$\sup_{L \geq L_0, i \in \{o,d\}} \mathbb{E} \left[\exp \left(\eta_1 \frac{U_N - Nu_i}{\sqrt{N}} \right) \middle| \mathcal{I}_{i,L} \right] < \infty \quad (\text{A.20})$$

for some fixed $\eta_1 > 0$ and sufficiently large fixed L_0 , where $\mathcal{I}_{o,L} = \mathcal{O}_L$ and $\mathcal{I}_{d,L} = \mathcal{D}_L$.

Proof. Introduce the bond fugacity parameter $\lambda = \log(e^\beta - 1)$; then $d\beta/d\lambda = p$. Put $B = |A|$ and $M = -U_N$, the number of monochromatic spin bonds. The partition function $e^{d\beta N} Z_{i,L}(\beta)$ is, up to the constant ordered multiplicity, a positive sum of $e^{\lambda B} q^{k(A)}$ over the fixed phase event. Therefore the first and second derivatives of its *logarithm*, $F_{i,L}(\beta) = \log[e^{d\beta N} Z_{i,L}(\beta)]$, with respect to λ are the conditional mean and variance of B .

Lemma A.1, the bounded derivatives of $\beta(\lambda)$, and the elementary $|F'_{i,L}| \leq CN$ give a second λ derivative bounded by CN throughout a window of width $\eta_2/\sqrt{N}$. At coexistence the first derivative is $-pNu_i + O(\sqrt{N})$, by (A.16). Taylor's theorem for log partition ratios at $\lambda_c \pm \eta_3/\sqrt{N}$ therefore proves

$$\mathbb{E} \left[e^{\eta_3 |B + pNu_i|/\sqrt{N}} \middle| \mathcal{I}_{i,L} \right] \leq C \quad (\text{A.21})$$

for a sufficiently small fixed $\eta_3 > 0$.

Conditional on the spin configuration, B is Binomial(M, p), since each monochromatic bond is retained independently with probability p . For a centered Bernoulli variable, the second derivative of its log moment-generating function is at most $1/4$; integrating twice bounds that function by $t^2/8$. Conditional independence therefore gives, for $D = B - pM$,

$$\mathbb{E}[e^{tD} \mid \sigma] \leq e^{dNt^2/8} \quad (t \in \mathbb{R}), \quad \mathbb{E}e^{t|D|/\sqrt{N}} \leq 2e^{dt^2/8}. \quad (\text{A.22})$$

This estimate remains bounded after conditioning on any event whose probability is bounded below, at the cost of dividing by that probability. At coexistence all cutoffs are inactive, and (A.9) with equal free energies, together with the tunneling bound, gives

$$P(\mathcal{O}_L) \rightarrow \frac{q}{q+1}, \quad P(\mathcal{D}_L) \rightarrow \frac{1}{q+1}, \quad P(\mathcal{T}_L) \leq Ce^{-bL^{d-1}}. \quad (\text{A.23})$$

Thus the conditional noise moments are uniformly bounded. Finally

$$|U_N - Nu_i| = |M + Nu_i| \leq p^{-1}\{|B + pNu_i| + |D|\}.$$

Cauchy–Schwarz, (A.21), and (A.22) prove (A.20) with $2\eta_1/p \leq \eta_3$. $\square$

This transfer uses a conditional binomial fluctuation estimate, not an identification of occupied bonds with spin energy. In particular, differentiating the restricted bond partition function gives the bond count first; the noise estimate is the additional step needed for the physical Hamiltonian.

Proof of Theorem 4.3. For necessity, the positive midpoint phase weights, (A.13), and $\sigma_i^2 \geq v_* > 0$ satisfy Theorem 3.4, with gap $N(u_d - u_o)$ and $s_N = \sqrt{N}$.

For sufficiency, use the positive decomposition by $\mathcal{O}_L, \mathcal{D}_L, \mathcal{T}_L$ on the Edwards–Sokal space. Lemma A.2 supplies the physical-energy moment bound. The exceptional mass is at most $Ce^{-bL^{d-1}}$, while the largest bath amplification is $\exp[O(N^2/c_N)]$. If $c_N \gg N^{3/2}$, its logarithm is $o(\sqrt{N}) = o(L^{d/2})$. Since $d-1 \geq d/2$ for $d \geq 2$, the exceptional contribution vanishes. Theorem 3.5 therefore gives total-variation convergence on the joint spin–bond space. The likelihood there is the exact function of $U_N(\sigma)$ in (2.3). Summing over bonds gives precisely the physical subsystem law, so projection proves the claimed spin conclusion. $\square$

The argument establishes a small fixed exponential moment on the $\sqrt{N}$ scale. It neither asserts a global Gaussian phase tail nor requires one. Optional independent quadratic momenta preserve the threshold: their centered Gamma transform adds the moment bound from Lemma 4.2, their mean shifts both centers by aN/β_c , and their variance adds a/β_c^2 to the weak phase limits. For $a > 0$ convolution also yields continuous local Gaussian energy limits, but no short-range pointwise density envelope is needed for Theorem 4.3.

B Exact Gaussian models and calibration geometry

Gaussian reservoirs and Gaussian descriptions of homogeneous coexistence peaks have a substantial history [7, 8]. Here they are exact probability models used to isolate calibration and phase-selection effects. Their interphase tails are not substituted for the microscopic laws of Section 4. A quadratic residual weight also differs from the exact power-law reservoir: the identification $\kappa = \beta^2/(C_B/k_B)$ describes a local entropy curvature, not a globally constant heat capacity for an exactly quadratic entropy.

B.1 Two scalar phases and the optimized error crossover

Let φ_{s^2} denote a centered normal density with standard deviation $s > 0$, and consider

$$p_N(E) = \frac{1}{2}\varphi_{s_N^2}(E - m_N) + \frac{1}{2}\varphi_{s_N^2}(E + m_N), \quad m_N/s_N \rightarrow \infty, \quad q_{N,t} \propto p_N e^{tE - \kappa_N E^2/2}, \quad \kappa_N \geq 0. \quad (\text{B.1})$$

The field $t \in \mathbb{R}$ is unrestricted. Completion of squares gives the transformed component variance, means, and upper-phase weight:

$$\tilde{s}_N^2 = \frac{s_N^2}{1 + \kappa_N s_N^2}, \quad \tilde{m}_{\pm,N} = \frac{\pm m_N + t s_N^2}{1 + \kappa_N s_N^2}, \quad v_{+,N}(t) = \frac{1}{1 + e^{-2tm_N/(1 + \kappa_N s_N^2)}}. \quad (\text{B.2})$$

Proposition B.1 (Scalar convergence and optimized crossover). *There exists a field sequence giving $d_{\text{TV}}(p_N, q_{N,t_N}) \rightarrow 0$ if and only if $\kappa_N m_N s_N \rightarrow 0$. If instead $\kappa_N m_N s_N \rightarrow a \in [0, \infty)$, then*

$$d_{\text{TV}}(p_N, q_{N,0}) \rightarrow 2\Phi(a/2) - 1, \quad (\text{B.3})$$

$$\inf_{t \in \mathbb{R}} d_{\text{TV}}(p_N, q_{N,t}) \rightarrow \min\{2\Phi(a/2) - 1, 1/2\}. \quad (\text{B.4})$$

Proof. Put $r_N = m_N/s_N$, $k_N = \kappa_N s_N^2$, and $b_N = t_N s_N$. In units of s_N , the transformed means are $(\pm r_N + b_N)/(1 + k_N)$ and their variance is $(1 + k_N)^{-1}$. If $\kappa_N m_N s_N \rightarrow 0$, then $k_N \rightarrow 0$ and the choice $t_N = 0$ preserves the weights exactly while making both standardized mean displacements vanish. Componentwise Gaussian TV convergence proves sufficiency.

For necessity, we use the scalar case of the label-matching argument given in full for Theorem B.2. Fix $R > 0$ and take the windows $|E \mp m_N| \leq R s_N$. Each has target mass tending to $\frac{1}{2}[2\Phi(R) - 1]$, and the windows are disjoint for large N . A transformed component has standard deviation $\tilde{s}_N \leq s_N$, so the smaller of its probabilities in these two windows tends to zero as $m_N/s_N \rightarrow \infty$. TV convergence therefore forces both components to keep weights with lower limit at least $\frac{1}{2}[2\Phi(R) - 1]$, each supplying one window, and letting $R \rightarrow \infty$ gives weights tending to 1/2 and centers $\tilde{m}_{\pm,N} = \pm m_N + O(s_N)$, matched in their natural order. Subtracting these center estimates and using the exact transformed means gives

$$\frac{2r_N}{1 + k_N} = 2r_N + O(1), \quad \frac{k_N r_N}{1 + k_N} = O(1).$$

Since $r_N \rightarrow \infty$, this forces $k_N \rightarrow 0$, so the variance ratio already tends to one. Restricting to the matched windows and then letting $R \rightarrow \infty$ gives TV convergence of each transformed component to its target normal. In particular, its probability above the target mean tends to 1/2. With standardized displacement $\delta_{\pm,N} = (b_N \mp k_N r_N)/(1 + k_N)$, this probability is $\Phi(\delta_{\pm,N}\sqrt{1 + k_N})$. Hence both displacements tend to zero. Subtracting them proves $k_N r_N \rightarrow 0$.

For the finite crossover, $k_N \rightarrow 0$ follows already from $k_N r_N \rightarrow a$ and $r_N \rightarrow \infty$. At $t_N = 0$ the two translated limiting normals have displacements $\mp a$ and unit variance. Separating their windows gives (B.3), using $d_{\text{TV}}(\mathcal{N}(0, 1), \mathcal{N}(a, 1)) = 2\Phi(|a|/2) - 1$.

To optimize, take any sequence of fields and a subsequence on which the transformed upper weight converges. If its limit lies strictly between zero and one, the logistic formula in (B.2) implies $b_N = O(1/r_N)$. The limiting component displacements are therefore still $\mp a$. The limiting TV is a convex function of the limiting upper weight, and reflection makes that function symmetric about $1/2$. It is minimized there, giving the first term in (B.4). If one weight tends to zero, a single bounded-width Gaussian cannot cover both target windows, so the limiting lower bound is $1/2$. This also covers means that escape all target windows. These subsequence bounds apply to arbitrary approximate minimizers of the infimum. The balanced choice attains the first bound. For $a > 0$, the choice $t_N = \kappa_N m_N$ centers the upper component exactly at m_N , makes its weight tend to one, and attains $1/2$. If $a = 0$, the balanced bound already vanishes. $\square$

The optimizer changes behavior at $a_* = 2\Phi^{-1}(3/4) \simeq 1.34898$. Above this value the TV objective can favor an accurate single phase while discarding the other one. An optimized scalar distance should therefore be accompanied by the phase probabilities.

Temperature calibration changes the required scale even within this simple model. Give the upper reference phase weight $w \neq 1/2$, with $0 < w < 1$, and hold the tangent of the quadratic bath at the canonical mean $\bar{E}_N = (2w - 1)m_N$. This means $q_N \propto p_N e^{-\kappa_N(E - \bar{E}_N)^2/2}$, or $t_N = \kappa_N \bar{E}_N$. The exact transformed phase odds are

$$\log \frac{v_{+,N}}{1 - v_{+,N}} = \log \frac{w}{1 - w} + \frac{2\kappa_N(2w - 1)m_N^2}{1 + \kappa_N s_N^2}. \quad (\text{B.5})$$

The same separated-window argument first requires $\kappa_N s_N^2 \rightarrow 0$; convergence of the weights then requires $\kappa_N m_N^2 \rightarrow 0$. Conversely, that condition makes both the weight and shape errors vanish. Thus for $m_N \asymp N$, $s_N \asymp \sqrt{N}$, this specified untuned prescription requires $\kappa_N N^2 \rightarrow 0$, whereas an appropriately tuned two-phase law needs only $\kappa_N N^{3/2} \rightarrow 0$. The reference point in this comparison is fixed by the target canonical mean, not by a self-consistent modified mean.

B.2 Several exchanged quantities and phase geometry

Let the Gaussian variable be $Y \in \mathbb{R}^d$, where here d is the number of exchanged quantities, not a spatial dimension. Fix distinct phase points $e_1, \dots, e_k$ and positive weights w_i summing to one. Set

$$p_N = \sum_{i=1}^k w_i \mathcal{N}(N e_i, N I_d), \quad q_{N,t} \propto p_N \exp(t \cdot Y - \kappa_N |Y|^2/2), \quad t \in \mathbb{R}^d, \quad \kappa_N \geq 0. \quad (\text{B.6})$$

With

$$s_N = (1 + \kappa_N N)^{-1}, \quad \alpha_N = \frac{\kappa_N N^2}{1 + \kappa_N N}, \quad z_N = \frac{N t}{1 + \kappa_N N}, \quad (\text{B.7})$$

the transformed covariance is $N s_N I_d$, its means are $N s_N(e_i + t)$, and its weights are

$$v_{i,N} = \frac{w_i \exp(z_N \cdot e_i - \alpha_N |e_i|^2/2)}{\sum_j w_j \exp(z_N \cdot e_j - \alpha_N |e_j|^2/2)}. \quad (\text{B.8})$$

For any affine dependence $(a_1, \dots, a_k)$, meaning $\sum_i a_i = 0$ and $\sum_i a_i e_i = 0$, this gives the exact invariant

$$\sum_i a_i \log(v_{i,N}/w_i) = -\frac{\alpha_N}{2} \sum_i a_i |e_i|^2. \quad (\text{B.9})$$

Every adjustable field has canceled.

Theorem B.2 (Geometry of Gaussian reservoir convergence). *For $k \geq 2$, some field sequence in (B.6) gives TV convergence if and only if*

$$\begin{cases} \kappa_N N^{3/2} \rightarrow 0, & \text{if the phase points are cospherical,} \\ \kappa_N N^2 \rightarrow 0, & \text{otherwise.} \end{cases} \quad (\text{B.10})$$

For $k = 1$, the condition is $\kappa_N N \rightarrow 0$. Cospherical means that the points lie on a common sphere of positive radius in $\mathbb{R}^d$; in $d = 1$ a sphere is a pair of points, so three distinct scalar phase points are never cospherical.

Proof. First justify matching the component labels. On the macroscopic variable Y/N , every transformed covariance is at most I_d/N . If TV tends to zero, each target phase neighborhood of positive mass must therefore receive a distinct transformed component of nonvanishing weight. There are exactly k components, so this is a bijection, and the matched transformed centers $s_N(e_i + t_N)$ lie in bounded neighborhoods of the phase points; in particular they are bounded. Along any subsequence, take a further subsequence on which s_N , the translation $s_N t_N$, and the matching permutation converge. The limiting map $e \mapsto se + a$ maps the finite phase set onto itself. Its positive diameter forces $s = 1$, and the arithmetic centroid then forces $a = 0$. Distinct phase points make the matching permutation the identity. It follows that $s_N \rightarrow 1$, $s_N t_N \rightarrow 0$, and $v_{i,N} \rightarrow w_i$ with the original labels.

Isolating each matched component in expanding fluctuation windows then gives its Gaussian TV convergence. In particular,

$$\sqrt{N}\{(s_N - 1)e_i + s_N t_N\} \rightarrow 0 \quad \text{for every } i. \quad (\text{B.11})$$

This necessity can also be detected by projection onto the direction of a component's mean displacement. Subtracting two equations in (B.11) proves $\kappa_N N^{3/2} \rightarrow 0$ for any two distinct phase points.

The phase-weight equations $v_i = w_i$, when $\alpha_N > 0$, are solvable exactly if and only if

$$|e_i|^2 = 2c \cdot e_i + b \quad \text{for every } i \quad (\text{B.12})$$

for some $c \in \mathbb{R}^d, b \in \mathbb{R}$. This is the sphere condition. Taking $t_N = \kappa_N N c$ then restores every weight exactly and makes the standardized mean changes $\kappa_N N^{3/2}(c - e_i)/(1 + \kappa_N N)$ tend to zero. This proves sufficiency in the spherical case.

If (B.12) has no solution, finite-dimensional linear algebra gives an affine dependence with $\sum_i a_i |e_i|^2 \neq 0$. Since $v_{i,N}/w_i \rightarrow 1$, (B.9) forces $\alpha_N \rightarrow 0$, equivalent to $\kappa_N N^2 \rightarrow 0$. Taking $t_N = 0$ proves sufficiency, including the component shapes. For a single phase, $t_N = \kappa_N N e_1$ aligns its mean exactly; matching the covariance is equivalent to $\kappa_N N \rightarrow 0$ and is necessary whatever mean is chosen. $\square$

Every affinely independent set of at most $d + 1$ points is cospherical, but larger sets may also be cospherical. The number of phases alone therefore does not determine the threshold in this vector model. For scalar phase points $-1, 0, 1$, the invariant reads

$$\frac{v_{+,N} v_{-,N}}{v_{0,N}^2} = \frac{w_+ w_-}{w_0^2} e^{-\alpha_N}. \quad (\text{B.13})$$

Three distinct scalar points are not cospherical, so their threshold is $\kappa_N N^2 \rightarrow 0$.

B.3 The exact optimum when some phases must disappear

An empty-sphere contact set is a nonempty subset of phase indices of the form

$$F(c) = \arg \min_{1 \leq i \leq k} |e_i - c|^2, \quad c \in \mathbb{R}^d. \quad (\text{B.14})$$

These are the contact sets of the usual lower-paraboloid, or Delaunay, construction [18]. Concentration of exponential families on faces also has an established general theory [15]; the theorem here determines the exact optimized TV loss under the specified diverging quadratic parameter. Only finitely many subsets can occur. Define $W_* := \max_c \sum_{i \in F(c)} w_i$.

Theorem B.3 (Optimal Gaussian phase loss). *For the model (B.6), if*

$$\kappa_N N^{3/2} \rightarrow 0, \quad \kappa_N N^2 \rightarrow \infty, \quad (\text{B.15})$$

then

$$\lim_{N \rightarrow \infty} \inf_{t \in \mathbb{R}^d} d_{\text{TV}}(p_N, q_{N,t}) = 1 - W_*. \quad (\text{B.16})$$

The optimization includes field sequences of unbounded norm.

Proof. For a fixed center c , choose $t_N = \kappa_N N c$. The phase score in (B.8) is, up to a common constant, $-\alpha_N |e_i - c|^2/2$. Since $\alpha_N \rightarrow \infty$, the weights converge to the reference weights conditioned on $F(c)$. The first condition in (B.15) makes every component shape change vanish in TV. This attains the upper bound $1 - \sum_{F(c)} w_i$, and maximizing over contact sets proves the desired upper bound.

For the lower bound, take any sequence of fields. First consider a subsequence with $|t_N| \geq \varepsilon > 0$. Put $u_N = t_N/|t_N|$ and $H_N = \max_i u_N \cdot e_i$. The weight outside indices with $u_N \cdot e_i \geq H_N - N^{-1/4}$ tends to zero: relative to a maximizer its linear score loses at least $c_1 N^{3/4}$ for a fixed $c_1 > 0$, whereas its quadratic score differs by only $O(\alpha_N) = o(\sqrt{N})$. For the remaining indices the projected transformed mean per particle is at least

$$s_N(H_N - N^{-1/4} + |t_N|) \geq H_N + \varepsilon/2$$

for all large N , uniformly also when $|t_N| \rightarrow \infty$. The target Gaussian mixture has negligible mass beyond $u_N \cdot Y/N > H_N + \varepsilon/4$, while the transformed law has mass tending to one there. Its TV therefore tends to one.

It remains to consider $t_N \rightarrow 0$. Pass to a subsequence on which all weights converge, and let I be the nonempty set of indices with positive limiting weights. The transformed means divided by N approach their own e_i , since $s_N \rightarrow 1$. Fixed disjoint small macroscopic neighborhoods of the phase points then imply

$$\liminf d_{\text{TV}}(p_N, q_{N,t_N}) \geq 1 - \sum_{i \in I} w_i. \quad (\text{B.17})$$

To constrain I , fix $i_0 \in I$ and put $c_N = z_N/\alpha_N$. The exact score ratios imply

$$\begin{aligned} c_N \cdot (e_i - e_{i_0}) - \frac{1}{2}(|e_i|^2 - |e_{i_0}|^2) &\rightarrow 0 & (i \in I), \\ c_N \cdot (e_j - e_{i_0}) - \frac{1}{2}(|e_j|^2 - |e_{i_0}|^2) &\leq o(1) & (1 \leq j \leq k). \end{aligned}$$

For the inequalities use $v_{i_0,N}$ bounded below and $v_{j,N} \leq 1$; the resulting log-ratio upper bound divided by α_N vanishes. These are approximate equalities and inequalities of a fixed finite linear system. Such a system cannot be infeasible while its right-hand side errors tend to zero. Indeed,

Farkas' alternative would supply fixed multipliers annihilating the unknown vector and yielding a strictly negative constant, contradicting the vanishing errors. Thus there is a finite vector c satisfying the limiting system exactly, even if c_N has no bounded subsequence. Its sphere contact set $F(c)$ contains I . Hence $\sum_{i \in I} w_i \leq W_*$. Combining with (B.17) proves the lower bound along every subsequence of approximate minimizers, and completes the proof. $\square$

For three scalar points $-1, 0, 1$, the largest possible retained sets are the two adjacent pairs. Thus the limiting optimal error is $\min(w_-, w_+)$, including $1/3$ for equal weights. The central phase cannot be the only discarded phase. More generally, Theorem B.3 identifies the exact retained-mass criterion, including the field sequences whose sphere centers escape to infinity; bounded-center analysis alone would supply only an upper bound.

B.4 Anisotropic curvature: the effective metric

For the same common covariance NI_d , replace $\kappa_N I_d$ by a symmetric positive-semidefinite matrix B_N . Completing squares gives

$$C_N = N(I_d + NB_N)^{-1}, \quad M_N = B_N(I_d + NB_N)^{-1}, \quad z_N = N(I_d + NB_N)^{-1}t_N, \quad (\text{B.18})$$

as the transformed covariance, effective metric, and effective field. The means are $N(I_d + NB_N)^{-1}(e_i + t_N)$, and $v_i \propto w_i \exp[z_N \cdot e_i - N^2 e_i^\top M_N e_i / 2]$. Accordingly the affine invariants and exact weight-restoration equations use M_N , not the raw matrix B_N . Completing the square for positive-definite M_N gives an ellipsoid, with a single point in the zero-radius case. For positive-rank singular M_N , a linear coefficient in its range gives an elliptic cylinder at positive radius or an affine subspace at zero radius. A nonzero linear component in its nullspace instead gives a paraboloid, possibly with additional flat directions. For $M_N = 0$, a nonzero linear coefficient gives a hyperplane; if the coefficient and compatible constant both vanish, the locus is the whole space. Inconsistent level constants give an empty locus in the completed-square or constant cases. These are finite- N algebraic statements. The scalar exponent classification above is not asserted for arbitrary anisotropic sequences or unequal phase covariance matrices.

C Interfacial tails and accuracy diagnostics

This appendix supplies the capillarity model behind the two-dimensional boundary range (6.17) and the exponent $\alpha = 1/2$ in the density criterion, and it records two diagnostic distinctions: histogram norms versus total variation, and barrier accuracy versus full-law accuracy.

C.1 A stated capillarity model and what it proves

The competition between background fluctuations and droplet surface costs is established in the work of Biskup, Chayes, and Kotecký [1]. The droplet-to-strip shape transition on a periodic square, at fixed excess volume fraction, was located by Leung and Zia [26] and analyzed for first-order simulations by Neuhaus and Hager [28]; droplet and strip energy regions in a toroidal two-dimensional Potts system are exhibited by Kim, Keyes, and Straub [22]. These results motivate the following capillarity model; none of them supplies this paper's assumed full density envelope or derives the square-torus rate used below for our microscopic systems.

To make the interfacial competition precise without importing an unproved microscopic rate function, consider the following model. Assume $Y_N = (E - e_{-,N})/\Delta_N$ is supported in a fixed compact interval K containing $[0, 1]$ and satisfies a large-deviation principle with speed $\sqrt{N}$ and

good rate I . The speed $\sqrt{N}$ is the two-dimensional surface scale $N^{(d-1)/d}$ at $d = 2$; the model is stated for that case, where the surface and fluctuation scales coincide. Assume I is continuous and finite on $[0, 1]$, infinite on $K \setminus [0, 1]$, zero exactly at $0, 1$, and positive on $(0, 1)$. The compact support is a convenient explicit hypothesis for this variational model, not an assertion about systems with unbounded kinetic energy.

Proposition C.1 (Capillarity variational law). *For $\Delta_N/N \rightarrow \ell > 0$ and $c_N/N^{3/2} \rightarrow \gamma > 0$, physical secant reweighting gives a large-deviation principle at speed $\sqrt{N}$ with rate*

$$I_\gamma(\theta) = I(\theta) - J_\gamma(\theta) - \min_{0 \leq u \leq 1} \{I(u) - J_\gamma(u)\}, \quad J_\gamma(\theta) = \frac{\beta^2 \ell^2}{2\gamma} \theta(1 - \theta). \quad (\text{C.1})$$

If

$$\gamma_* = \frac{\beta^2 \ell^2}{2} \sup_{0 < \theta < 1} \frac{\theta(1 - \theta)}{I(\theta)} < \infty, \quad (\text{C.2})$$

then $\gamma > \gamma_*$ leaves only the endpoints as rate minimizers. For $\gamma < \gamma_*$ all minimizing energies lie away from the endpoints and $d_{\text{TV}}(P_N, Q_N) \rightarrow 1$.

Proof. Equation (6.14), expanded on the fixed compact K , gives $h_N/\sqrt{N} \rightarrow J_\gamma$ uniformly. The cutoff lies beyond K for large N because $c_N/\Delta_N \rightarrow \infty$. For a continuous bounded tilt on a compact space, the large-deviation upper bound applied to a finite cover of sets where the tilt varies by at most ϵ , and the lower bound on an open neighborhood of each point, give

$$\lim_N \frac{1}{\sqrt{N}} \log \mathbb{E} e^{h_N} = \max_{\theta \in K} \{J_\gamma(\theta) - I(\theta)\}.$$

The same bounds for restricted integrals prove (C.1). If $\gamma > \gamma_*$, $I(\theta) > J_\gamma(\theta)$ throughout the interior. If $\gamma < \gamma_*$, the minimum of $I - J_\gamma$ is strictly negative, whereas its endpoint values are zero. Compactness and continuity place its minimizing set a positive distance from the endpoints. A neighborhood of that set has reservoir probability tending to one and canonical probability tending to zero, proving TV. $\square$

For $\gamma > \gamma_*$ the proposition specifies macroscopic energy concentration, not full-state TV or the fluctuation law within a phase. At equality, a rate function alone cannot determine weights among tied minimizers. Multiplication by arbitrary positive constants, or factors with logarithm $o(\sqrt{N})$, near tied minima leaves the same large-deviation rate and can change their weights or select a single minimum. This is nonidentifiability from the stated model input, rather than a missing consequence of the rate principle.

For an explicit square-torus capillarity model, set

$$I(\theta) = \tau \min\{2\sqrt{\pi\theta}, 2, 2\sqrt{\pi(1-\theta)}\}, \quad 0 \leq \theta \leq 1, \quad \tau > 0. \quad (\text{C.3})$$

The three prescribed branches represent an isotropic circular droplet, a slab with two interfaces, and the complementary bubble. This is the isotropic-tension form of the standard droplet, strip, and bubble construction on a square torus [26, 28]; its droplet-to-strip crossover is at $\theta = 1/\pi$. Equation (C.3) defines the variational model; it is not a derived Wulff law for the lattice Potts model.

Corollary C.2 (Square-torus minimizers). *For (C.3), $\gamma_* = \beta^2 \ell^2 / (16\tau)$. Above γ_* the only minimizers are $0, 1$; below it the unique minimizer is $1/2$; at equality the minimizers are $0, 1/2, 1$.*

Proof. For all θ ,

$$I(\theta) \geq 8\tau\theta(1 - \theta), \quad (\text{C.4})$$

with equality exactly at $0, 1/2, 1$. The slab branch obeys this because $\theta(1 - \theta) \leq 1/4$. For the droplet branch, away from $\theta = 0$ the desired bound follows from $8\sqrt{\theta(1 - \theta)} \leq 16/(3\sqrt{3}) < 2\sqrt{\pi}$; the bubble branch is symmetric. Put $A = \beta^2\ell^2/(2\gamma)$. If $A < 8\tau$, only endpoints minimize $I - A\theta(1 - \theta)$. If $A = 8\tau$, equality in (C.4) identifies the three minimizers. If $A > 8\tau$, using $I(1/2) = 2\tau$ gives

$$[I(\theta) - A\theta(1 - \theta)] - [I(1/2) - A/4] \geq (A - 8\tau)(\theta - 1/2)^2,$$

which is strictly positive off $1/2$. $\square$

C.2 Histogram norms can hide a persistent error

The distinction between TV and an unscaled density norm is visible even in one phase. Let p_σ be the density of $\mathcal{N}(0, \sigma^2)$ and q_σ that of $\mathcal{N}(0, r\sigma^2)$, for fixed $0 < r < 1$. Direct Gaussian integration gives

$$\|p_\sigma - q_\sigma\|_2^2 = \frac{1}{2\sqrt{\pi}\sigma} \left[1 + r^{-1/2} - 2\sqrt{\frac{2}{1+r}} \right] \rightarrow 0 \quad (\sigma \rightarrow \infty), \quad (\text{C.5})$$

although

$$d_{\text{TV}}(p_\sigma, q_\sigma) = 2 \left[\Phi\left(\sqrt{\frac{\log r}{r-1}}\right) - \Phi\left(\sqrt{\frac{r \log r}{r-1}}\right) \right] > 0. \quad (\text{C.6})$$

For the latter formula, the two densities cross at $|E|/\sigma = \sqrt{r \log r / (r - 1)}$; integrating their difference inside the crossings gives (C.6). For histogram probabilities in bins of fixed width $a > 0$, a rescaled Riemann sum gives $\sum_j (P_j - Q_j)^2 \sim a \|p_\sigma - q_\sigma\|_2^2$. Consequently its Euclidean norm decays as $\sigma^{-1/2}$, or $N^{-1/4}$ when $\sigma \asymp \sqrt{N}$. The normal laws still have a fixed variance ratio and fixed TV error. This is a normalization effect of the chosen norm. It is the distinction promised in Section 1: the finite-bath Potts comparison of Griffin, Matty, and Swendsen measures agreement by the unweighted Euclidean norm of energy-probability differences, their equation (23), and reports its decay with size [20]. The example shows that such decay is compatible with a fixed within-phase variance ratio and a fixed full-state error. Gaussian reservoir fluctuation formulas themselves are established [7].

C.3 Barrier accuracy requires a separate scale

Suppose canonical and reservoir energy densities are positive at specified energies e_- and $E^\dagger$, and define their fixed-energy log barriers by $B_P = \log[p(e_-)/p(E^\dagger)]$ and $B_Q = \log[q(e_-)/q(E^\dagger)]$. With $h(e_-) = 0$, normalization cancels exactly:

$$B_Q - B_P = -h(E^\dagger). \quad (\text{C.7})$$

The same identity holds for ratios of point probabilities in a discrete energy law. For a quadratic endpoint-balanced weight with gap $\Delta = 2m$, the midpoint gain is $\kappa m^2/2 = \kappa \Delta^2/8$. For the exact physical secant bath,

$$h\left(\frac{e_- + e_+}{2}\right) = c \log \cosh\left(\frac{\beta \Delta}{2c}\right). \quad (\text{C.8})$$

It follows directly from (6.14) at $\theta = 1/2$, and is positive.

If $c \gg \Delta$, the lowering is $\beta^2 \Delta^2 / (8c) [1 + O((\Delta/c)^2)]$. More generally, for $\Delta \rightarrow \infty$ and inverse temperature bounded away from zero and infinity, its vanishing is equivalent to $c \gg \Delta^2$. Indeed,

$\log \cosh x$ is bounded below by a positive constant times $\min(x^2, |x|)$ and above by $x^2/2$. The lower bound follows from continuity of the corresponding ratios on $0 < |x| \leq 1$ and $|x| \geq 1$, with positive limits at zero and infinity. Thus a subsequence with Δ/c bounded below has barrier error of order at least Δ , while in the other regime it is comparable to Δ^2/c .

For an extensive gap, absolute midpoint-barrier accuracy therefore requires $c \gg N^2$, even though the full-state two-phase TV error can vanish for $c \gg N^{3/2}$. If a specified canonical barrier has size $B_P \asymp N^{(d-1)/d}$ and $c \gg N$, relative accuracy of this fixed-energy barrier instead requires $c \gg N^2/B_P \asymp N^{1+1/d}$. These statements concern density ratios at specified energies. They do not locate shifted peaks or saddle points, nor do they constitute a theorem about a dynamical transition rate.

D Shared-bath extensions: fixed phase counts and linear capacity

This appendix extends Section 5 in two directions under the same hypotheses (5.1)–(5.2). The first selects a prescribed number of plus-phase copies among a fixed number $m \geq 2$ of copies. The second treats the centered reservoir at capacity proportional to N , where the bath also correlates the fluctuations inside an opposite-phase pair and the marginals acquire a computable finite error.

D.1 Selecting a fixed phase count among finitely many copies

For a fixed integer $m \geq 2$, take the reference law $\Pi_N^{(m)} = P_N^{\otimes m}$. Let K_N count the copies in the plus phase and fix $k \in \{0, \dots, m\}$. Center the reservoir at

$$M_{m,k,N} = (m-k)e_{-,N} + ke_{+,N}, \quad \mathcal{E}_N = M_{m,k,N} + c_N/\beta_N. \quad (\text{D.1})$$

Denote the reweighted m -copy law by $Q_N^{[m]}$ and its one-copy marginals by $Q_{j,N}^{[m]}$. Write $f = k/m$ and

$$p_k = \binom{m}{k} w_+^k w_-^{m-k}, \quad H(f) = -f \log f - (1-f) \log(1-f), \quad (\text{D.2})$$

with the usual zero-term convention at $f = 0, 1$.

Proposition D.1 (Fixed-count selection and total correlation). *Under the hypotheses of Theorem 5.1,*

$$d_{\text{TV}}(Q_N^{[m]}, \Pi_N^{(m)}(\cdot \mid K_N = k)) \rightarrow 0, \quad (\text{D.3})$$

$$d_{\text{TV}}(Q_N^{[m]}, \Pi_N^{(m)}) \rightarrow 1 - p_k, \quad (\text{D.4})$$

$$d_{\text{TV}}(Q_{j,N}^{[m]}, P_N) \rightarrow |f - w_+|, \quad (\text{D.5})$$

$$D_{\text{KL}} \left(Q_N^{[m]} \left\| \bigotimes_{j=1}^m Q_{j,N}^{[m]} \right. \right) \rightarrow mH(f) - \log \binom{m}{k}. \quad (\text{D.6})$$

Proof. On $K_N = k$, the energy deviation from $M_{m,k,N}$ is tight on scale $\sqrt{N}$ because m is fixed. On any other count it equals $(K_N - k)\Delta_N + O_{\Pi_N^{(m)}}(\sqrt{N})$ conditionally. The proof of Theorem 5.1 therefore gives L^1 convergence of the bounded weight to $\mathbf{1}_{\{K_N=k\}}$, proving (D.3). All phase assignments with count k have the same canonical probability, so conditioning selects them uniformly. Each marginal has plus-phase probability f and the unchanged phase-conditional microscopic laws. This proves the two total-variation formulas.

The bounded-likelihood argument from Proposition 5.2 gives $D_{\text{KL}}(Q_N^{[m]} \parallel \Pi_N^{(m)}) \rightarrow -\log p_k$ and

$$D_{\text{KL}}(Q_{j,N}^{[m]} \parallel P_N) \rightarrow f \log \frac{f}{w_+} + (1-f) \log \frac{1-f}{w_-}.$$

Subtracting the m marginal divergences by the relative-entropy chain rule gives (D.6). This argument also covers $k = 0, m$, since $r \log r$ is continuous at zero. $\square$

When $w_+ = k/m$ with $0 < k < m$, every complete subsystem marginal becomes canonical and the limiting total correlation equals $-\log p_k$. For $m = 2, k = 1$ it is one bit. At $k = 0$ or m , the selected phase assignment is unique and the limiting total correlation is zero. There is no statement here for a number of copies increasing with N ; the phase probability and normalization bounds used in the proof are not uniform in m .

D.2 The centered reservoir at linear capacity

At c_N proportional to N , the bath also correlates fluctuations inside an opposite-phase assignment. Retain the centered calibration (5.3), and now strengthen the phase assumption to

$$\frac{E_N - e_{i,N}}{\sqrt{N}} \mid A_{i,N} \Rightarrow \mathcal{N}(0, v_i), \quad v_- > 0, \quad v_+ > 0, \quad c_N/N \rightarrow \gamma \in (0, \infty). \quad (\text{D.7})$$

Define

$$\alpha = \beta^2/\gamma, \quad V = v_- + v_+, \quad A_\gamma = (1 + \alpha V)^{-1/2}, \quad p_O = 2w_-w_+. \quad (\text{D.8})$$

The following results concern this specified calibration. They make no claim of optimality over all possible total-energy choices at linear capacity.

Theorem D.2 (Gaussian fluctuation tilt at linear capacity). *Under (5.1)–(5.2) and (D.7), the two opposite-phase assignments each have limiting probability 1/2 under Q_N , and the same-phase assignments disappear. Within an opposite assignment, order the standardized energies by phase. Their limiting law is centered Gaussian with covariance*

$$\Sigma_\gamma = D - \frac{\alpha}{1 + \alpha V} v v^\top, \quad D = \text{diag}(v_-, v_+), \quad v = (v_-, v_+)^\top. \quad (\text{D.9})$$

The one-copy variances and squared correlation coefficient of this limiting Gaussian law are

$$\hat{v}_i = \frac{v_i(1 + \alpha v_j)}{1 + \alpha V} \quad (j \neq i), \quad \rho_\gamma^2 = \frac{\alpha^2 v_- v_+}{(1 + \alpha v_-)(1 + \alpha v_+)}. \quad (\text{D.10})$$

In particular $0 < \hat{v}_i < v_i$, and the correlation is negative.

Proof. For a state in phase i write $Y_{i,N} = (E_N - e_{i,N})/\sqrt{N}$. On an opposite assignment, the exact weight (5.4) converges uniformly on compact standardized windows to

$$W_\gamma(y_-, y_+) = \exp[-\alpha(y_- + y_+)^2/2]. \quad (\text{D.11})$$

On a same-phase assignment its argument $x_N = \beta_N T_N / c_N$ tends to the nonzero constant $\pm \beta \ell / \gamma$. If that constant is negative, the strict inequality $x + \log(1 - x) < 0$ makes the log weight at most a negative constant times N on each compact window. For a positive constant, the cutoff gives zero whenever $x_N \geq 1$, and otherwise $x_N + \log(1 - x_N) \leq -x_N^2/2$. This also gives suppression, including the case where the limiting center meets the cutoff. Thus no expansion around zero is used for the same-phase assignments at linear capacity.

Conditional weak convergence, tightness, and $0 \leq W_N \leq 1$ imply

$$z_N \longrightarrow p_O A_\gamma, \quad A_\gamma = \int W_\gamma(y_-, y_+) \varphi_{v_-}(y_-) \varphi_{v_+}(y_+) dy_- dy_+. \quad (\text{D.12})$$

The integral equals $(1 + \alpha V)^{-1/2}$ because the canonical sum has variance V . Exchange symmetry makes the two surviving assignment weights equal. Multiplication by (D.11) adds $\alpha \mathbf{1}\mathbf{1}^\top$ to the Gaussian precision matrix D^{-1} . Inverting this matrix gives (D.9) ; its entries give (D.10) . $\square$

The covariance conclusion is weak convergence of the standardized energy laws. It is not total-variation convergence of microscopic phase laws to Gaussian laws. A stronger and well-defined microscopic approximation is available: retain the original microscopic reference Π_N , replace W_N by $\mathbf{1}_{O_N} W_\gamma(Y_{S_1, N}, Y_{S_2, N})$, and normalize. The resulting finite- N law approaches Q_N in full total variation, since the two bounded unnormalized weights approach each other in $L^1(\Pi_N)$. Gaussian laws enter only in the limiting evaluation of bounded energy integrals.

Proposition D.3 (Full marginal error at linear capacity). *Under the same assumptions, either complete microscopic marginal has the limit*

$$\lim_{N \rightarrow \infty} d_{\text{TV}}(Q_N^{(j)}, P_N) = \frac{1}{2} \sum_{i \in \{-, +\}} \int_{\mathbb{R}} \left| \frac{1}{2} \varphi_{\hat{v}_i}(y) - w_i \varphi_{v_i}(y) \right| dy. \quad (\text{D.13})$$

For balanced weights this is strictly positive and equals

$$\sum_{i \in \{-, +\}} \left[\Phi\left(\frac{x_i}{\sqrt{\hat{v}_i}}\right) - \Phi\left(\frac{x_i}{\sqrt{v_i}}\right) \right], \quad x_i^2 = \frac{\hat{v}_i v_i \log(v_i / \hat{v}_i)}{v_i - \hat{v}_i}, \quad (\text{D.14})$$

where Φ is the standard normal CDF.

Proof. On a phase- i window, integrate the kernel against the other copy's energy law. Its opposite-phase contribution converges, uniformly on compact y -intervals, to $w_j g_i(y)$, where

$$g_i(y) = \frac{1}{\sqrt{1 + \alpha v_j}} \exp\left[-\frac{\alpha y^2}{2(1 + \alpha v_j)}\right], \quad \frac{g_i(y)}{A_\gamma} = \frac{\varphi_{\hat{v}_i}(y)}{\varphi_{v_i}(y)}. \quad (\text{D.15})$$

To justify uniformity, first restrict the other standardized energy to a fixed compact interval, where the kernel convergence is uniform and its continuous limit is uniformly continuous. Weak convergence then gives uniformity in y by a finite grid argument. Tightness and the bound by one control the discarded other-copy tail. The same-phase contribution vanishes by the same argument and the cutoff estimate in Theorem D.2.

Thus the exact marginal likelihood tends on compact phase windows to $g_i(y)/(2w_i A_\gamma)$. It is globally bounded by $1/z_N$. Tightness therefore controls the rest of the marginal TV integral, and weak Gaussian convergence gives

$$\frac{1}{2} \sum_i w_i \int \left| \frac{g_i(y)}{2w_i A_\gamma} - 1 \right| \varphi_{v_i}(y) dy,$$

which is (D.13) . At balance it is one half the sum of the two Gaussian total-variation distances. The two centered Gaussian densities cross at $\pm x_i$; integration between the crossings gives (D.14) . $\square$

For completeness, the joint error also has an explicit limit. The same bounded likelihood argument gives

$$\lim d_{\text{TV}}(Q_N, \Pi_N) = \frac{1 - p_O}{2} + \frac{p_O}{2} \mathbb{E} \left| \frac{e^{-\alpha G^2/2}}{p_O A_\gamma} - 1 \right|, \quad G \sim \mathcal{N}(0, V). \quad (\text{D.16})$$

Equivalently, if $t_\gamma^2 = -2 \log(p_O A_\gamma)/\alpha$, which is positive because $p_O \leq 1/2$ and $A_\gamma < 1$, this limit is

$$2\Phi \left(t_\gamma \sqrt{\frac{1 + \alpha V}{V}} \right) - 1 - p_O \left[2\Phi \left(\frac{t_\gamma}{\sqrt{V}} \right) - 1 \right].$$

This follows by integrating over the positive-likelihood set $|G| < t_\gamma$, using variance $V/(1 + \alpha V)$ under the tilted law.

Proposition D.4 (Phase and fluctuation information at linear capacity). *Under the hypotheses of Theorem D.2,*

$$\lim_{N \rightarrow \infty} I_{Q_N}(X_1; X_2) = \log 2 - \frac{1}{2} \log(1 - \rho_\gamma^2) = \log 2 + \frac{1}{2} \log \frac{(1 + \alpha v_-)(1 + \alpha v_+)}{1 + \alpha V}. \quad (\text{D.17})$$

The second term is strictly positive and is independent of w_-, w_+ .

Proof. Boundedness and continuity of $x \log x$ on $[0, 1]$ allow exactly the same compact-window argument as for (D.12). Under the Gaussian tilt the sum of standardized energies has variance $V/(1 + \alpha V)$. Therefore

$$\lim D_{\text{KL}}(Q_N \| \Pi_N) = -\log p_O + \frac{1}{2} \log(1 + \alpha V) - \frac{\alpha V}{2(1 + \alpha V)}. \quad (\text{D.18})$$

For the marginals, the compact convergence of their likelihoods in Proposition D.3, their uniform bound by $1/z_N$, and uniform continuity of $r \log r$ give

$$\lim D_{\text{KL}}(Q_N^{(j)} \| P_N) = \frac{1}{2} \sum_i \log \frac{1}{2w_i} + \frac{1}{2} \sum_i D_{\text{KL}}(\mathcal{N}(0, \hat{v}_i) \| \mathcal{N}(0, v_i)). \quad (\text{D.19})$$

These are statements about bounded microscopic likelihood integrals, not differential-entropy convergence. The Gaussian divergence is $\frac{1}{2}[\hat{v}_i/v_i - 1 - \log(\hat{v}_i/v_i)]$. Substituting this expression and (D.18) into the finite-divergence chain rule (5.12) gives (D.17). $\square$

The first term in (D.17) is the phase assignment information. The second comes from correlated energy fluctuations inside the opposite phases. As the coefficient γ increases, these formulas approach the one-bit and balanced canonical marginal limits of (5.5). This is a comparison of the displayed limiting formulas; the proved intermediate-window result already treats sequences with $c_N/N \rightarrow \infty$ directly.

The positive variance hypotheses hold for the plain short-range Potts models by (A.13)–(A.14), also after the balancing shift. They hold for the mean-field benchmark when $a > 0$. The pure-spin mean-field model has $v_+ = 0$ and is outside the positive-variance linear-capacity statements above, while it remains covered by the opposite-phase and fixed-count selection theorems.

References

- [1] Marek Biskup, Lincoln Chayes, and Roman Kotecký. On the formation/dissolution of equilibrium droplets. *Europhysics Letters*, 60(1):21–27, 2002. URL: <https://arxiv.org/abs/math-ph/0207012>, [arXiv:math-ph/0207012](https://arxiv.org/abs/math-ph/0207012), [doi:10.1209/epl/i2002-00312-y](https://doi.org/10.1209/epl/i2002-00312-y).
- [2] Christian Borgs, Jennifer Chayes, Tyler Helmuth, Will Perkins, and Prasad Tetali. Efficient sampling and counting algorithms for the Potts model on $\mathbb{Z}^d$ at all temperatures. <https://arxiv.org/abs/1909.09298>, 2022. [arXiv:1909.09298v3](https://arxiv.org/abs/1909.09298). URL: <https://arxiv.org/abs/1909.09298>, [arXiv:1909.09298](https://arxiv.org/abs/1909.09298).
- [3] Christian Borgs, Jennifer T. Chayes, and Prasad Tetali. Tight bounds for mixing of the Swendsen–Wang algorithm at the Potts transition point. *Probability Theory and Related Fields*, 152:509–557, 2012. URL: <https://arxiv.org/abs/1011.3058>, [arXiv:1011.3058](https://arxiv.org/abs/1011.3058).
- [4] Christian Borgs, Roman Kotecký, and Salvador Miracle-Solé. Finite-size scaling for Potts models. *Journal of Statistical Physics*, 62:529–551, 1991. URL: <https://www.microsoft.com/en-us/research/wp-content/uploads/2016/11/Finite-Size-Scaling-for-Potts-Models.pdf>, [doi:10.1007/BF01017971](https://doi.org/10.1007/BF01017971).
- [5] Michele Campisi. On the limiting cases of nonextensive thermostatics. *Physics Letters A*, 366:335–338, 2007. URL: <https://arxiv.org/abs/cond-mat/0611068>, [arXiv:cond-mat/0611068](https://arxiv.org/abs/cond-mat/0611068), [doi:10.1016/j.physleta.2007.01.082](https://doi.org/10.1016/j.physleta.2007.01.082).
- [6] Murty S. S. Challa and J. H. Hetherington. Gaussian ensemble: An alternate Monte Carlo scheme. *Physical Review A*, 38:6324–6337, 1988. URL: <https://harvest.aps.org/v2/journals/articles/10.1103/PhysRevA.38.6324/fulltext>, [doi:10.1103/PhysRevA.38.6324](https://doi.org/10.1103/PhysRevA.38.6324).
- [7] Murty S. S. Challa and J. H. Hetherington. Gaussian ensemble as an interpolating ensemble. *Physical Review Letters*, 60:77–80, 1988. URL: <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.60.77>, [doi:10.1103/PhysRevLett.60.77](https://doi.org/10.1103/PhysRevLett.60.77).
- [8] Murty S. S. Challa, D. P. Landau, and K. Binder. Finite-size effects at temperature-driven first-order transitions. *Physical Review B*, 34:1841–1852, 1986. URL: <https://journals.aps.org/prb/abstract/10.1103/PhysRevB.34.1841>, [doi:10.1103/PhysRevB.34.1841](https://doi.org/10.1103/PhysRevB.34.1841).
- [9] Murty S. S. Challa, D. P. Landau, and K. Binder. Monte Carlo studies of finite-size effects at first-order transitions. *Phase Transitions*, 24–26:343–369, 1990. URL: <https://www.tandfonline.com/doi/abs/10.1080/01411599008210236>, [doi:10.1080/01411599008210236](https://doi.org/10.1080/01411599008210236).
- [10] Or Cohen, Vladimir Rittenberg, and Tridib Sadhu. Shared information in classical mean-field models. *Journal of Physics A: Mathematical and Theoretical*, 48:055002, 2015. URL: <https://arxiv.org/abs/1409.5520>, [arXiv:1409.5520](https://arxiv.org/abs/1409.5520), [doi:10.1088/1751-8113/48/5/055002](https://doi.org/10.1088/1751-8113/48/5/055002).
- [11] David S. Corti, Donya Ohadi, Ricardo Fariello, and Mark J. Uline. Microcanonical thermodynamics of small ideal gas systems. *The Journal of Physical Chemistry B*,

- 127(15):3431–3442, 2023. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10123661/>, doi:10.1021/acs.jpcb.3c00455.
- [12] Marius Costeniuc, Richard S. Ellis, and Hugo Touchette. Complete analysis of phase transitions and ensemble equivalence for the Curie–Weiss–Potts model. *Journal of Mathematical Physics*, 46:063301, 2005. URL: <https://arxiv.org/abs/cond-mat/0410744>, arXiv:cond-mat/0410744, doi:10.1063/1.1904507.
 - [13] Marius Costeniuc, Richard S. Ellis, and Hugo Touchette. Nonconcave entropies from generalized canonical ensembles. *Physical Review E*, 74:010105, 2006. URL: <https://arxiv.org/abs/cond-mat/0605213>, arXiv:cond-mat/0605213, doi:10.1103/PhysRevE.74.010105.
 - [14] Marius Costeniuc, Richard S. Ellis, Hugo Touchette, and Bruce Turkington. The generalized canonical ensemble and its universal equivalence with the microcanonical ensemble. *Journal of Statistical Physics*, 119:1283–1329, 2005. URL: <https://arxiv.org/abs/cond-mat/0408681>, arXiv:cond-mat/0408681, doi:10.1007/s10955-005-4407-0.
 - [15] Imre Csiszár and František Matúš. Closures of exponential families. *The Annals of Probability*, 33(2):582–600, 2005. URL: <https://arxiv.org/abs/math/0503653>, arXiv:math/0503653, doi:10.1214/009117904000000766.
 - [16] Persi Diaconis and David A. Freedman. Conditional limit theorems for exponential families and finite versions of de Finetti’s theorem. *Journal of Theoretical Probability*, 1:381–410, 1988. URL: <https://digicoll.lib.berkeley.edu/record/86140>, doi:10.1007/BF01048727.
 - [17] Jörn Dunkel and Stefan Hilbert. Consistent thermostats forbids negative absolute temperatures. *Nature Physics*, 10:67–72, 2014. URL: <https://arxiv.org/abs/1304.2066>, arXiv:1304.2066, doi:10.1038/nphys2815.
 - [18] Herbert Edelsbrunner and Raimund Seidel. Voronoi diagrams and arrangements. *Discrete & Computational Geometry*, 1:25–44, 1986. URL: <https://pub.ista.ac.at/~edels/Papers/1986-11-VoronoiArrangements.pdf>, doi:10.1007/BF02187681.
 - [19] Daniel Gandolfo, Jean Ruiz, and Marc Wouts. Limit theorems and coexistence probabilities for the Curie–Weiss Potts model with an external field. *Stochastic Processes and their Applications*, 120:84–104, 2010. URL: <https://arxiv.org/abs/0811.2735>, arXiv:0811.2735, doi:10.1016/j.spa.2009.10.011.
 - [20] William Griffin, Michael Matty, and Robert H. Swendsen. Finite thermal reservoirs and the canonical distribution. *Physica A*, 484:1–10, 2017. URL: <https://arxiv.org/abs/1608.05455>, arXiv:1608.05455, doi:10.1016/j.physa.2017.04.143.
 - [21] Stefan Hilbert, Peter Hänggi, and Jörn Dunkel. Thermodynamic laws in isolated systems. *Physical Review E*, 90:062116, 2014. URL: <https://arxiv.org/abs/1408.5382>, arXiv:1408.5382, doi:10.1103/PhysRevE.90.062116.
 - [22] Jaegil Kim, Thomas Keyes, and John E. Straub. Communication: Iteration-free, weighted histogram analysis method in terms of intensive variables. *The Journal of Chemical Physics*, 135(6):061103, 2011. URL:

- <https://people.bu.edu/straub/pdf/pubs/JCP.135.061103.2011.pdf>,
doi:10.1063/1.3626150.
- [23] Roman Kotecký and Senya B. Shlosman. First-order phase transitions in large entropy lattice models. *Communications in Mathematical Physics*, 83:493–515, 1982. URL: <https://projecteuclid.org/euclid.cmp/1103920966>, doi:10.1007/BF01208713.
 - [24] Lahoussine Laanait, Alain Messenger, Salvador Miracle-Solé, Jean Ruiz, and Senya Shlosman. Interfaces in the Potts model I: Pirogov–Sinai theory of the Fortuin–Kasteleyn representation. *Communications in Mathematical Physics*, 140:81–91, 1991. URL: <https://projecteuclid.org/euclid.cmp/1104247908>.
 - [25] Joel L. Lebowitz, Jerome K. Percus, and Loup Verlet. Ensemble dependence of fluctuations with application to machine computations. *Physical Review*, 153:250–254, 1967. URL: <https://harvest.aps.org/v2/journals/articles/10.1103/PhysRev.153.250/fulltext>, doi:10.1103/PhysRev.153.250.
 - [26] K. Leung and R. K. P. Zia. Geometrically induced transitions between equilibrium crystal shapes. *Journal of Physics A: Mathematical and General*, 23(20):4593–4602, 1990. URL: <https://iopscience.iop.org/article/10.1088/0305-4470/23/20/021>, doi:10.1088/0305-4470/23/20/021.
 - [27] Yuri Mishin. Thermodynamic theory of equilibrium fluctuations. *Annals of Physics*, 363:48–97, 2015. URL: <https://arxiv.org/abs/1507.05662>, arXiv:1507.05662, doi:10.1016/j.aop.2015.09.015.
 - [28] Thomas Neuhaus and Johannes S. Hager. 2D crystal shapes, droplet condensation, and exponential slowing down in simulations of first-order phase transitions. *Journal of Statistical Physics*, 113:47–83, 2003. URL: <https://arxiv.org/abs/cond-mat/0201324>, arXiv:cond-mat/0201324, doi:10.1023/A:1025718703965.
 - [29] A. Ramírez-Hernández, H. Larralde, and F. Leyvraz. Violation of the zeroth law of thermodynamics in systems with negative specific heat. *Physical Review Letters*, 100:120601, 2008. URL: <https://arxiv.org/abs/0802.1748>, arXiv:0802.1748, doi:10.1103/PhysRevLett.100.120601.
 - [30] Arnau Riera, Christian Gogolin, and Jens Eisert. Thermalization in nature and on a quantum computer. *Physical Review Letters*, 108:080402, 2012. URL: <https://arxiv.org/abs/1102.2389>, arXiv:1102.2389, doi:10.1103/PhysRevLett.108.080402.
 - [31] Yasushi Yoneta and Akira Shimizu. Squeezed ensemble for systems with first-order phase transitions. *Physical Review B*, 99:144105, 2019. URL: <https://arxiv.org/abs/1903.04111>, arXiv:1903.04111, doi:10.1103/PhysRevB.99.144105.